



Aakash

Medical | IIT-JEE | Foundations



शुभ दीपावली

**HOME
ASSIGNMENT**

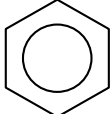
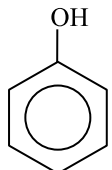
KOTA

“Apna Aakash” Road No. 1, IPIA, Kota (Raj.)

[CLICK HERE TO GET AAKASH TEST PAPERS](#)

IMPORTANT REAGENTS & CHEMICAL TESTS

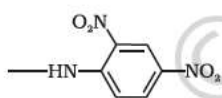
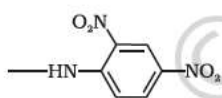
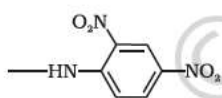
HYDROCARBON		
1.	Controlled oxidation	$\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Cu}/523\text{K}/100\text{atm}} \text{CH}_3\text{CHO}$ $\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mo}_2\text{O}_3} \text{CH}_3\text{COOH}$ $\text{CH}_4 + \text{O}_2 \xrightarrow{(\text{CH}_3\text{COO})_2\text{Mn}} \text{CH}_3\text{COOH}$ $(\text{CH}_3)_3\text{CH} \xrightarrow{\text{KMnO}_4} (\text{CH}_3)_2\text{C=O}$
2.	Anhydrous AlCl_3 :	Lewis acid used as a catalyst. $\bullet \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \xrightarrow[\text{Isomerization}]{\text{AlCl}_3/\text{Conc.HCl}} \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ $\bullet \text{CH}_2 = \text{CH}_2 + \text{CH}_3\text{COCl} \xrightarrow{\text{AlCl}_3} \text{CH}_3 - \text{CH}(\text{CH}_3) - \text{COCl}$ $\bullet \text{C}_6\text{H}_6 + \text{CH}_3\text{Cl} \xrightarrow[\text{Friedel Craft's alkylation}]{\text{AlCl}_3} \text{C}_6\text{H}_5\text{CH}_3$ $\bullet \text{C}_6\text{H}_6 + \text{CH}_3\text{COCl} \xrightarrow[\text{Friedel Craft's acetylation}]{\text{AlCl}_3} \text{C}_6\text{H}_5\text{COCH}_3$ $\bullet n - \text{Hexane} \xrightarrow{\text{AlCl}_3} \text{Benzene}$ R-COOH, RNO_2 , R-NH_2 do not give F.C.R.
3.	Lindlar's catalyst	Pd/CaCO_3 $\text{R} - \text{C} \equiv \text{C} - \text{R} + \text{H}_2 \xrightarrow[\text{Pd/C}]{\text{Pd/CaCO}_3} \text{R} - \text{CH} = \text{CH} - \text{R}$
4.	$\text{Na} + \text{Liq.NH}_3$	$\text{R} - \text{C} \equiv \text{C} - \text{R} + \text{H}_2 \xrightarrow{\text{Na} + \text{Liq.NH}_3} \text{R} - \text{CH} = \text{CH} - \text{R}$
5.	Alc. KOH and NaNH_2	$\bullet \begin{array}{c} > \text{C} - \text{C} < \\ \quad \\ \text{H} \quad \text{Br} \end{array} \xrightarrow[\text{(-HBr)}]{\text{Elimination}} \begin{array}{c} > \text{C} = \text{C} < \\ \quad \\ \text{H} \quad \text{Br} \end{array}$ $\bullet \begin{array}{c} \text{R} - \text{C} - \text{C} - \text{R} \\ \quad \\ \text{H} \quad \text{Br} \end{array} \xrightarrow{(-2\text{HBr})} \begin{array}{c} \text{R} - \text{C} = \text{C} - \text{R} \\ \quad \\ \text{H} \quad \text{Br} \end{array}$ (Gem dihalide) $\bullet \begin{array}{c} \text{R} - \text{CH} - \text{CH} - \text{R} \\ \quad \\ \text{Br} \quad \text{Br} \end{array} \xrightarrow{(-2\text{HBr})} \begin{array}{c} \text{R} - \text{C} = \text{C} - \text{R} \\ \quad \\ \text{Br} \quad \text{Br} \end{array}$ (Vicinal dihalide)
6.	Zinc metal :	$\bullet \text{C}_6\text{H}_5\text{OH} \xrightarrow[\text{deoxygenation}]{\text{Zn}} \text{C}_6\text{H}_6$ $\bullet 2\text{R} - \text{X} \xrightarrow{\text{Zn}} \text{R} - \text{R}$ $\bullet \begin{array}{c} \text{CH}_2 - \text{CH}_2 \\ \quad \\ \text{Cl} \quad \text{Cl} \end{array} \xrightarrow{\text{Zn}} \text{CH}_2 = \text{CH}_2$ $\bullet 2 \text{CH}_3 - \text{CH} \begin{array}{l} \text{Cl} \\ \text{Cl} \end{array} \xrightarrow{\text{Zn}} \text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ $\bullet \begin{array}{c} \text{Br} \quad \text{Br} \\ \quad \\ \text{R} - \text{C} - \text{C} - \text{R} \\ \quad \\ \text{Br} \quad \text{Br} \end{array} \xrightarrow{2\text{Zn}} \text{R} - \text{C} = \text{C} - \text{R}$
7.	Sulphuric acid: (conc. H_2SO_4)	Order of dehydration of alcohol $3^\circ > 2^\circ > 1^\circ$

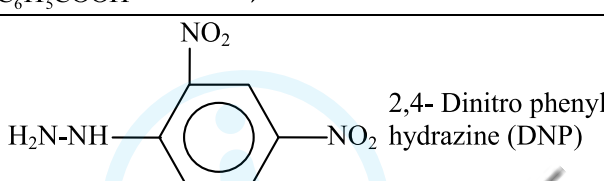
		$\text{C}_2\text{H}_5\text{OH} \xrightarrow{170^\circ\text{C}} \text{CH}_2 = \text{CH}_2$ $\xrightarrow{140^\circ\text{C}} \text{C}_2\text{H}_5 - \text{O} - \text{C}_2\text{H}_5$ $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3 \xrightarrow{85\% \text{H}_3\text{PO}_4 \& 440\text{K}} \text{CH}_3\text{CH}=\text{CH}_2$ $\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)(\text{CH}_3) \xrightarrow{20\% \text{H}_3\text{PO}_4 \& 358\text{K}} \text{CH}_3\text{C}(\text{CH}_3)=\text{CH}_2$  $\xrightarrow{\text{Sulphonation}}$
8.	Bromine (Br_2) :	Test of unsaturation $\text{>C}=\text{C}< + \text{Br}_2 \xrightarrow{\text{CCl}_4} \text{Red}$ $\text{--C}\equiv\text{C--} + \text{Br}_2 \xrightarrow{\text{CCl}_4} \text{Red}$  + $\text{Br}_2 \xrightarrow{\text{H}_2\text{O}} \text{Red}$
9.	Addition of HX	$\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HBr/HCl/HI}}$ $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow[\text{Peroxide}]{\text{HBr}}$ $\text{CH}_3 - \text{C}\equiv\text{CH} \xrightarrow{\text{HBr}} \text{A} \xrightarrow{\text{HBr}} \text{B}$
10.	Baeyer's Reagent:	1% cold alkaline dil KMnO_4 (Pink Colour) Test of unsaturation $\text{>C}=\text{C}< \longrightarrow$ $\text{--C}\equiv\text{C--} \longrightarrow$ Benzene $\longrightarrow$ No reaction
11.	H^+ and KMnO_4	$(\text{CH}_3)_2\text{C}=\text{CH}_2 \longrightarrow$ $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3 \longrightarrow$
12.	$\text{Ni/Pd/Pt} + \text{H}_2$	$\text{CH}_2 = \text{CH}_2 \longrightarrow$ $\text{HC}\equiv\text{CH} \longrightarrow$ $\text{R}-\text{CHO} \longrightarrow$ $\text{R}_2\text{CO} \longrightarrow$ $\text{RCOOH} \longrightarrow$ $\text{RCOOR}^1 \longrightarrow$ $\text{R}(\text{CO})\text{O}(\text{CO})\text{R}' \longrightarrow$ $\text{R}-\text{C}\equiv\text{N} \longrightarrow$ $\text{RCONH}_2 \longrightarrow$ $\text{R}-\text{NC} \longrightarrow$
13.	Hydrating agents	$\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{CH} = \text{CH}_2 \xrightarrow{\text{Dil. H}_2\text{SO}_4}$ $\xrightarrow[\text{(ii) H}_2\text{O}_2/\text{OH}^-]{\text{(i) B}_2\text{H}_6}$ $\xrightarrow[\text{(ii) NaBH}_4/\text{OH}^-]{\text{(i) (AcO)}_2\text{Hg/H}_2\text{O}}$
14.	Sodium metal:	$2\text{R}-\text{X} \xrightarrow[\text{Wurtz reaction}]{\text{Na/dry ether}}$ $\text{ROH} \xrightarrow{\text{Na}}$ $\text{R}-\text{C}\equiv\text{CH} \xrightarrow{\text{Na}}$

		$\bullet \text{PhOH} \xrightarrow{\text{Na}}$ $\bullet \text{RCOOH} \xrightarrow{\text{Na}}$
15.	Sodalime: (NaOH + CaO)	$\text{RCOOH or RCOONa} \xrightarrow[\Delta]{\text{decarboxylation}}$ $\text{PhCOONa} \xrightarrow[\Delta]{\text{decarboxylation}}$
16.	Cyclic Polymerisation	$3\text{CH} \equiv \text{CH} \xrightarrow[\text{Fe tube}]{\text{Red Hot}}$ $2\text{CH} \equiv \text{CH} + \text{CH}_3\text{C} \equiv \text{CH} \longrightarrow$ $1\text{CH} \equiv \text{CH} + 2\text{CH}_3\text{C} \equiv \text{CH} \longrightarrow$ $3\text{CH}_3\text{C} \equiv \text{CH} \longrightarrow$
17.	Miscellaneous	$\text{C}_6\text{H}_6 \xrightarrow[\text{Dark, cold}]{\text{Anhy. AlCl}_3 + \text{X}_2 \text{ or Fe} + \text{X}_2}$ $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow[\text{Dark, cold}]{\text{Anhy. AlCl}_3 + \text{X}_2 \text{ or Fe} + \text{X}_2} \rightarrow \text{Major}$ $\text{C}_6\text{H}_6 \xrightarrow[500\text{K}]{\text{Cl}_2 + \text{UV}}$ $\text{C}_6\text{H}_6 \xrightarrow{\text{H}_2 / \text{Ni}}$
HALOALKANES & HALOARENES		
18.	Lucas Reagent :	Conc. HCl/anhy. ZnCl ₂ Test of alcohols $\text{R} - \text{OH} \rightarrow \text{R} - \text{Cl}$ (white turbidity) Reactivity order $3^\circ \text{ROH} > 2^\circ \text{ROH} > 1^\circ \text{ROH}$ $3^\circ \text{R} - \text{OH} \xrightarrow{\text{HCl}}$ $2^\circ \text{R} - \text{OH} \xrightarrow{\text{HCl}} \text{no reaction}$ $1^\circ \text{R} - \text{OH} \xrightarrow{\text{HCl}} \text{No reaction}$ $1^\circ, 2^\circ, 3^\circ \text{R} - \text{OH} \xrightarrow[\text{ZnCl}_2]{\text{HCl}}$
19.	PCl ₅ :	$\bullet \text{R} - \text{OH} \xrightarrow{\text{PCl}_5}$ $\bullet >\text{C} = \text{O} \xrightarrow{\text{PCl}_5}$ $\bullet \text{R} - \text{COOH} \xrightarrow{\text{PCl}_5}$ $\bullet \text{R} - \text{C} - \text{Z} \xrightarrow{\text{PCl}_5}$ $\quad \quad \quad \parallel$ $\quad \quad \quad \text{O}$ (Acid derivative) $\bullet \text{R} - \text{CONH}_2 \xrightarrow{\text{dehydration}}$ $\text{R} - \text{OH} \xrightarrow{\text{PBr}_3 / \text{PCl}_3}$ $\text{R} - \text{OH} \xrightarrow{\text{NaBr} + \text{H}_2\text{SO}_4}$ $\text{R} - \text{OH} \xrightarrow{\text{Red P} + \text{X}_2 (\text{X} = \text{Br}, \text{I})}$ $\text{RCOOH} \xrightarrow{\text{PCl}_3}$
20.	SOCl ₂ :	$\bullet \text{R} - \text{OH} \longrightarrow$ $\bullet \text{R} - \text{COOH} \longrightarrow$
21.	KCN/AgCN	$\bullet \text{R} - \text{X} \xrightarrow{\text{KCN}}$ $\bullet \text{R} - \text{X} \xrightarrow{\text{AgCN}}$
22.	KNO ₂ /AgNO ₂	$\bullet \text{R} - \text{X} \xrightarrow{\text{KNO}_2}$ $\bullet \text{R} - \text{X} \xrightarrow{\text{AgNO}_2}$
23.	ArSN	$\text{C}_6\text{H}_5\text{Cl} \xrightarrow[(\text{ii}) \text{H}^+]{(\text{i}) \text{NaOH}, 623\text{K}, 300\text{atm}}$ 2,4,6 – trinitrochlorobenzene $\xrightarrow[\text{H}_2\text{O}]{\text{Warm}}$
24.	ESR	$\text{C}_6\text{H}_5\text{Cl} \xrightarrow{\text{Cl}_2 + \text{Anhyd. FeCl}_3}$ $\text{C}_6\text{H}_6 \xrightarrow{\text{Cl}_2 + \text{Anhyd. FeCl}_3}$ $\text{C}_6\text{H}_5\text{Cl} \xrightarrow{\text{HNO}_3 + \text{Conc. H}_2\text{SO}_4}$

		$\text{C}_6\text{H}_6 \xrightarrow{\text{HNO}_3 + \text{Conc. H}_2\text{SO}_4} \text{C}_6\text{H}_5\text{Cl} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ $\text{C}_6\text{H}_5\text{Cl} + \text{CH}_3\text{Cl} \xrightarrow{\text{Anh. AlCl}_3} \text{major}$ $\text{C}_6\text{H}_5\text{Cl} + \text{CH}_3\text{COCl} \xrightarrow{\text{Anh. AlCl}_3} \text{major}$
25.	Silver metal :	$2\text{CHCl}_3 + 6\text{Ag} \longrightarrow \text{HC} \equiv \text{CH} + 6\text{AgI}$
26.	Frankland's Reagent:	$\text{R}'\text{X} \xrightarrow{\text{R}_2\text{Zn}} \text{R} - \text{R}' + \text{RZnX}$
27.	$\text{R}'\text{COOAg}$	$\text{R} - \text{X} \longrightarrow$
28.	N- Bromo succinimide (NBS)	$\bullet \text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{allylic bromination}} \text{Br} - \text{CH}_2 - \text{CH} = \text{CH}_2$ $\bullet \text{C}_6\text{H}_5 \xrightarrow{\text{Benzylic bromination}} \text{C}_6\text{H}_5\text{CH}_2\text{Br}$
ALCOHOL, PHENOL & ETHER		
29.	$\text{LiAlH}_4/\text{H}^\oplus$:	Acts as a reducing agent. $\bullet \text{R} - \text{CHO} \longrightarrow$ $\bullet \text{R} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{R} \longrightarrow$ $\bullet \text{R} - \text{COOH} \longrightarrow$ $\bullet \text{R} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{Z} \longrightarrow$ (Acid derivative) $\bullet \text{R} - \text{CONH}_2 \longrightarrow$ $\bullet \text{R} - \text{C} \equiv \text{N} \longrightarrow$ $\bullet \text{R} - \text{NO}_2 \longrightarrow$ $\bullet \text{R} - \text{X} \longrightarrow$ $> \text{C} = \text{C} <$ does not reduce.
30.	NaBH_4	$\bullet \text{R} - \text{CHO} \longrightarrow$ $\bullet \text{R} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{R} \longrightarrow$ $\bullet \text{R} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{Cl} \longrightarrow$
31.	Red P/HI	Acts as a reducing agent $\bullet \text{R} - \text{CH}_2\text{OH} \longrightarrow$ $\bullet \text{R} - \text{CHO} \longrightarrow$ $\bullet \text{R}_2\text{CO} \longrightarrow$ $\bullet \text{R} - \text{COOH} \longrightarrow$
32.	Grignard's Reagent:	RMgX (alkyl Magnesium halide) $\bullet \text{Active-H} \xrightarrow{\text{RMgX}} \text{R} - \text{H}$ Containing compounds (H attached with 3 or more EN elements) $\bullet \text{R} - \text{Br} + \text{Mg} \xrightarrow{\text{dry ether}}$ $\bullet \text{R}' - \text{X} \xrightarrow{\text{RMgX}} \text{R} - \text{R}'$ $\bullet \text{H} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH} \xrightarrow{(i) \text{R}'\text{MgX} (ii) \text{H}_3\text{O}^\oplus}$

		$\bullet \text{R}-\text{CH} \xrightarrow{\text{(i) R'MgX (ii) H}_3\text{O}^{\oplus}} \text{O}$ $\bullet \text{R}-\text{C}-\text{R} \xrightarrow{\text{(i) R'MgX (ii) H}_3\text{O}^{\oplus}} \text{O}$ $\bullet \text{R}-\text{C}-\text{Z} \xrightarrow{\text{R'MgX}} \text{A} \xrightarrow{\text{R'MgX}} \text{B}$ O (Acid derivatives) $\bullet \text{R}-\text{C} \equiv \text{N} \xrightarrow{\text{(i) RMgX (ii) H}_3\text{O}^{\oplus}}$ $\bullet \text{RMgX} \xrightarrow{\text{(1) CO}_2 / \text{Dry ether} \rightarrow \text{(2) H}_3\text{O}^+}$
33.	Miscellaneous	$\text{C}_6\text{H}_6 \xrightarrow{\text{Oleum}} \text{A} \xrightarrow{\text{(i) NaOH (ii) H}^+} \text{B}$ $\text{C}_6\text{H}_5-\text{CH}(\text{CH}_3)_2 \xrightarrow{\text{O}_2} \text{A} \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \text{B} + \text{C}$ Cumene $\text{C}_6\text{H}_5\text{OH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Na}_2\text{C}_2\text{O}_7}$ $\text{CO} + \text{H}_2 \xrightarrow[\text{High P, High temp}]{\text{ZnO}-\text{Cr}_2\text{O}_3}$
34.	Na, Al, NaOH, NaHCO ₃	$\text{R}-\text{OH} \xrightarrow{\text{Na}} \quad \text{Ph}-\text{OH} \xrightarrow{\text{Na}} \quad \text{RCOOH} \xrightarrow{\text{Na}}$ $\text{R}-\text{OH} \xrightarrow{\text{NaOH}} \quad \text{Ph}-\text{OH} \xrightarrow{\text{NaOH}} \quad \text{RCOOH} \xrightarrow{\text{NaOH}}$ $\text{R}-\text{OH} \xrightarrow{\text{NaHCO}_3} \quad \text{Ph}-\text{OH} \xrightarrow{\text{NaHCO}_3} \quad \text{RCOOH} \xrightarrow{\text{NaHCO}_3}$ $\text{t-butyl alcohol} \xrightarrow{\text{Al}}$
35.	Copper metal :	$\bullet \text{R}-\text{CH}_2\text{OH} \xrightarrow[\text{dehydrogenation}]{\text{Cu}/300^\circ\text{C}}$ $\bullet \text{R}_2\text{CHOH} \xrightarrow[\text{dehydrogenation}]{\text{Cu}/300^\circ\text{C}}$ $\bullet (\text{CH}_3)_3\text{C}-\text{OH} \xrightarrow[\text{dehydration}]{\text{Cu}/300^\circ\text{C}}$
36.	Pyridinium chloro chromate: (PCC)	 $\text{R}-\text{CH}_2\text{OH} \xrightarrow{\text{PCC}} \quad (\text{Best yield})$
37.	Jones Reagent	$\text{CrO}_3 \text{ aq. H}_2\text{SO}_4 \quad \text{RCH}_2\text{OH} \xrightarrow[\text{H}_2\text{O/Aceto}]{\text{CrO}_3/\text{H}_2\text{SO}_4}$ $\text{R}_2\text{CHOH} \xrightarrow{\text{CrO}_3/\text{H}_2\text{SO}_4}$
38.	Ferric chloride: (FeCl ₃)	Test of Phenol: Phenol $\longrightarrow$ Violet colour
39.	ESR	$\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{dil. HNO}_3} \text{Major}$ $\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{conc. HNO}_3} \text{Major}$ $\text{C}_6\text{H}_5\text{OH} \xrightarrow[273\text{K}]{\text{Br}_2 \text{ in CS}_2} \text{Major}$ $\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{Br}_2 \text{ water}} \text{Major}$ $3\text{-methylphenol} \xrightarrow{\text{Dinitration}}$ $\text{Phenylmethanoate} \xrightarrow{\text{Mononitration}}$ $\text{C}_6\text{H}_5\text{OCH}_3 \xrightarrow[\text{ethanoic acid}]{\text{Br}_2 \text{ in}} \text{Major}$ $\text{C}_6\text{H}_5\text{OCH}_3 \xrightarrow[\text{CS}_2]{\text{CH}_2\text{Cl}_2 + \text{Anhyd. AlCl}_3} \text{Major}$

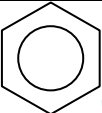
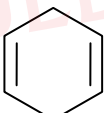
		$\text{C}_6\text{H}_5\text{OCH}_3 \xrightarrow[\text{CS}_2]{\text{CH}_3\text{COCl} + \text{Anhyd. AlCl}_3} \text{Major}$ $\text{C}_6\text{H}_5\text{OCH}_3 \xrightarrow[\text{HNO}_3]{\text{H}_2\text{SO}_4} \text{Major}$																																
40.	Cleavage of C-O bond in ethers	$\text{R}-\text{O}-\text{R}' \xrightarrow{\text{HX}} \text{R}-\text{X} + \text{R}'-\text{OH}$ $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3 \xrightarrow{\text{HI}} \text{A} + \text{B}$ $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2-\text{O}-\text{C}_2\text{H}_5 \xrightarrow{\text{HI}} \text{A} + \text{B}$ $\text{CH}_3\text{CH}_2\text{CH}_2-\text{O}-\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_3 \xrightarrow{\text{HI}} \text{A} + \text{B}$ $\text{Ph}-\text{CH}_2-\text{O}-\text{Ph} \xrightarrow{\text{HI}} \text{A} + \text{B}$																																
ALDEHYDE, KETONE AND CARBOXYLIC ACID																																		
41.	DIBAL-H/AlH(i-Bu) ₂	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2\text{CH}_2-\text{CN} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) DIBAL-H}} \rightarrow$ $\text{CH}_3(\text{CH}_2)_9-\text{COOC}_2\text{H}_5 \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) DIBAL-H}} \rightarrow$																																
42.	CrO ₃ + (CH ₃ CO) ₂ O	$\text{C}_6\text{H}_5\text{CH}_3 + \text{CrO}_3 + (\text{CH}_3\text{CO})_2\text{O} \xrightarrow{0-10^\circ\text{C}} \text{A} \xrightarrow{\text{H}_3\text{O}^+/\Delta} \text{B}$																																
43.	Cl ₂ /hν	$\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow{\text{Cl}_2/h\nu} \text{A} \xrightarrow{\text{H}_2\text{O}} \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow{\text{Cl}_2/h\nu} \text{A} \xrightarrow{\text{H}_2\text{O}} \text{C}_6\text{H}_5\text{CHO}$ $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow{\text{Cl}_2/h\nu} \text{A} \xrightarrow{\text{H}_2\text{O}} \text{C}_6\text{H}_5\text{COOH}$ $\text{C}_6\text{H}_5\text{CH}_3 \xrightarrow{\text{Cl}_2/h\nu} \text{A} \xrightarrow{\text{FeBr}_3/\text{Br}_2} \text{B} \xrightarrow{\text{Zn}+\text{HCl}} \text{C}$																																
44.	R ₂ Cd	$2\text{R MgX} \xrightarrow{\text{CdCl}_2} \text{A} \xrightarrow{\text{R}'\text{COCl}} \text{B}$ $2\text{CH}_3\text{MgX} \xrightarrow{\text{CdCl}_2} \text{A} \xrightarrow{\text{C}_2\text{H}_5\text{COCl}} \text{B}$																																
45.	ROH & HCl	$\text{RCHO} \xrightarrow[\text{HCl gas}]{\text{R}'\text{OH}} \text{A} \xrightarrow[\text{HCl gas}]{\text{R}'\text{OH}} \text{B} \xrightarrow{\text{Aq. HCl}} \text{C}$ $\text{RCOR} + \text{Ethylene Glycol} \xrightarrow{\text{HCl gas}} \text{A} \xrightarrow{\text{dil. HCl}} \text{B}$																																
46.	Addition then elimination of Carbonyl Compound	$\text{>C=O} + \text{H}_2\text{N-Z} \rightleftharpoons \left[\text{>C} \begin{matrix} \text{OH} \\ \text{NHZ} \end{matrix} \right] \rightarrow \text{>C=N-Z} + \text{H}_2\text{O}$ <table><thead><tr><th>Z</th><th>Reagent name</th><th>Carbonyl derivative</th><th>Product name</th></tr></thead><tbody><tr><td>-H</td><td>Ammonia</td><td></td><td></td></tr><tr><td>-R</td><td>Amine</td><td></td><td></td></tr><tr><td>-OH</td><td>Hydroxylamine</td><td></td><td></td></tr><tr><td>-NH₂</td><td>Hydrazine</td><td></td><td></td></tr><tr><td>-HN-</td><td>Phenylhydrazine</td><td></td><td></td></tr><tr><td>-HN-</td><td>2,4-Dinitrophenylhydrazine</td><td></td><td></td></tr><tr><td>-NH-C(=O)-NH₂</td><td>Semicarbazide</td><td></td><td></td></tr></tbody></table>	Z	Reagent name	Carbonyl derivative	Product name	-H	Ammonia			-R	Amine			-OH	Hydroxylamine			-NH ₂	Hydrazine			-HN- 	Phenylhydrazine			-HN- 	2,4-Dinitrophenylhydrazine			-NH-C(=O)-NH ₂	Semicarbazide		
Z	Reagent name	Carbonyl derivative	Product name																															
-H	Ammonia																																	
-R	Amine																																	
-OH	Hydroxylamine																																	
-NH ₂	Hydrazine																																	
-HN- 	Phenylhydrazine																																	
-HN- 	2,4-Dinitrophenylhydrazine																																	
-NH-C(=O)-NH ₂	Semicarbazide																																	
47.	KMnO ₄ , Chromic Acid	$\text{C}_6\text{H}_5-\text{CH}- \& \text{C}_6\text{H}_5-\text{CO}- \xrightarrow{\text{KMnO}_4/\text{OH}^-} \text{A} \xrightarrow{\text{H}_3\text{O}^+} \text{B}$ $\text{PhCH}_2\text{CH}_3 \xrightarrow{\text{H}^+/\text{KMnO}_4} \rightarrow$																																
48.	Miscellaneous	$\text{R}-\text{C}\equiv\text{N} \xrightarrow[\text{H}_2\text{O}]{\text{H}^+ \text{ or } \text{OH}^-} \rightarrow$ $\text{R}-\text{CONH}_2 \xrightarrow[\Delta]{\text{H}_3\text{O}^+} \rightarrow$ $\text{PhCOOH} + \text{NH}_3 \xrightarrow{\Delta} \text{A} \xrightarrow{\Delta} \text{B}$																																

		Phthalic acid + $\text{NH}_3 \xrightarrow{\Delta} \text{A} \xrightarrow{\text{Strong Heating}} \text{B} \xrightarrow{\Delta} \text{C}$
49.	Esterification	$\text{RCOOH} + \text{R}'\text{OH} \xrightarrow{\text{H}^+} \text{A} + \text{H}_2\text{O}$
50.	B_2H_6 & H^+	$\text{R}-\text{COOH} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) B}_2\text{H}_6}$ B_2H_6 does not reduce ester, nitro & halo compounds
51.	Sodium bisulphate: (NaHSO_3)	It is used for test of carbonyl compounds. $\text{>C=O} \longrightarrow \text{>C} \begin{matrix} \text{OH} \\ \text{SO}_3\text{Na} \end{matrix}$ (White crystal) $\bullet \text{PhCOPh} \longrightarrow \text{No reaction}$ $\bullet \text{CH}_3\text{COPh} \longrightarrow \text{No reaction}$
52.	E.S.R.	$\text{C}_6\text{H}_5\text{COOH} \xrightarrow{\text{Conc. HNO}_3 + \text{H}_2\text{SO}_4}$ $\text{C}_6\text{H}_5\text{CHO} \xrightarrow{\text{Conc. HNO}_3 + \text{H}_2\text{SO}_4}$ $\text{C}_6\text{H}_5\text{COOH} \xrightarrow{\text{Br}_2 + \text{FeBr}_3}$
53.	Brady's Reagent :	 2,4- Dinitro phenyl hydrazine (DNP) DNP test for carbonyl compounds $\text{>C=O} \xrightarrow[\text{(-H}_2\text{O)}]{\text{DNP}} \text{>C=N-NH-} \text{C}_6\text{H}_3(\text{NO}_2)_2$ (Yellow orange crystal)
54.	Tollen's Reagent:	$(\text{NH}_4\text{OH} + \text{AgNO}_3)$ $\bullet \text{RCHO} \longrightarrow \text{RCOO}^- + \text{Ag} \downarrow$ (Silver mirror) jtr niZ.k $\bullet \text{PhCHO} \longrightarrow \text{PhCOO}^- + \text{Ag} \downarrow$ (Silver mirror) jtr niZ.k $\bullet \text{R}-\text{C}\equiv\text{CH} \longrightarrow \text{R}-\text{C}\equiv\text{C Ag}$ (White ppt) α - Hydroxy Keto, Hemiacetal, PhNHOH , HCOOH Also give silver Mirror test Ketone, Ketal, $\text{R}-\text{C}\equiv\text{C}-\text{R}$, RCOOH does not silver mirror test
55.	Schiff's Reagent:	It is a pink coloured organic dye (Magenta or p-rosaniline hydrochloride) which has been decolourised by passing SO_2 gas through it. $\bullet \text{R}-\text{CHO} \longrightarrow \text{Pink Colour (restored)}$ $\bullet \text{R}-\text{C}(=\text{O})-\text{R} \longrightarrow \text{No pink colour}$
56.	Fehling's Solution:	$\text{Aq. CuSO}_4 + \text{Rochelle's salt (sodium potassium tartarate)}$ $\text{RCHO} \longrightarrow \text{RCOO}^- + \text{Cu}_2\text{O (Red ppt)}$ α - Hydroxy Keto, Hemiacetal also gives fehling test Ph-CHO RCOR does not give fehling test
57.	Benedict's Solution:	$\text{Aq. CuSO}_4 + \text{Sodium citrate} + \text{Na}_2\text{CO}_3$ $\text{RCHO} \longrightarrow \text{RCOO}^- + \text{Cu}_2\text{O (Red ppt)}$ α - Hydroxy Keto, Hemiacetal also gives fehling test Ph-CHO RCOR does not give fehling test
AMINE		
58.	P_2O_5 :	$\bullet 2\text{CH}_3\text{COOH} \xrightarrow[\Delta]{\text{dehydration}} (\text{CH}_3\text{CO})_2\text{O}$ $\bullet \text{CH}_3\text{CONH}_2 \xrightarrow[\Delta]{\text{dehydration}} \text{CH}_3\text{CN}$
59.	Reduction	$\text{C}_6\text{H}_5\text{NO}_2 \xrightarrow[\text{Or Fe+HCl}]{\text{H}_2 + \text{Pd or Sn} + \text{HCl}}$

		$\text{RCN} \xrightarrow[\text{C}_2\text{H}_5\text{OH}]{\text{Na(Hg)}}$
60.	Amonolysis	$\text{NH}_3 + \text{R} - \text{X}(\text{excess}) \longrightarrow$ $\text{NH}_3(\text{excess}) + \text{R} - \text{X} \longrightarrow$ $\text{C}_6\text{H}_5\text{CH}_2\text{Cl} \xrightarrow{\text{NH}_3} \text{A} \xrightarrow{2\text{CH}_3\text{Cl}} \text{B}$
61.	Hinsberg's Reagent:	$\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ (Benzene sulphonyl chloride) It is used to distinguish between 1°/2°/3° amines. • $\text{R} - \text{NH}_2 \xrightarrow[\text{(ii) KOH}]{\text{(i) C}_6\text{H}_5\text{SO}_2\text{Cl}}$ Soluble salt • $\text{R}_2\text{NH} \xrightarrow[\text{(ii) KOH}]{\text{(i) C}_6\text{H}_5\text{SO}_2\text{Cl}}$ Insoluble • $\text{R}_3\text{N} \xrightarrow{\text{C}_6\text{H}_5\text{SO}_2\text{Cl}}$ No reaction
62.	Nitrous acid: (HNO_2)	It is used to distinguish between 1°/2°/3° amines. • $\text{R} - \text{NH}_2 \xrightarrow{\text{HNO}_2} \text{R} - \text{OH} + \text{N}_2 \uparrow + \text{H}_2\text{O}$ • $\text{R}_2\text{NH} \xrightarrow{\text{HNO}_2} \text{R}_2\text{N} \cdot \text{NO} + \text{H}_2\text{O}$ (N-nitroso compound) • $\text{R}_3\text{N} \xrightarrow[0^\circ\text{C}]{\text{HNO}_2} \text{R}_3\text{N} \cdot \text{HNO}_2$ (salt) • $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[0^\circ-5^\circ\text{C}]{\text{HNO}_2} \text{C}_6\text{H}_5\text{N}_2\text{Cl}$ (Benzene diazonium chloride)
63.	Miscellaneous	$\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[\text{or Ethanoic anhydride}]{\text{CH}_3\text{COCl}} \text{A} \xrightarrow[\text{CH}_3\text{COOH}]{\text{Br}_2} \text{B} \xrightarrow{\text{H}^+ \text{ or } \text{OH}^-} \text{C}$ $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[\text{Pyridine}]{\text{CH}_3\text{COCl}} \text{A} \xrightarrow{\text{HNO}_3 + \text{H}_2\text{SO}_4} \text{B} \xrightarrow{\text{H}^+ \text{ or } \text{H}^-} \text{C}$
64.	E.S.R.	$\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{Br}_2 \text{ water}} \longrightarrow$ $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{HNO}_3 + \text{H}_2\text{SO}_4} \longrightarrow$ $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{H}_2\text{SO}_4} \text{A} \xrightarrow{453-473\text{K}} \text{B}(\text{Zwitter ion})$
65.	Reaction of Benznediazonium salt	$\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu}_2\text{Cl}_2/\text{HCl}} \text{A}$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu}_2\text{Br}_2/\text{HBr}} \text{B}$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{CuCN/KCN}} \text{C}$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu/HCl}} \text{D}$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{Cu/HBr}} \text{E}$ A, B, C are Example of Sandmeyer Reaction D & E are Example of Gatterman Reaction $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{KI}} \longrightarrow$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{H}_2\text{O}} \longrightarrow$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{H}_3\text{PO}_2 + \text{H}_2\text{O} \text{ or } \text{C}_2\text{H}_5\text{OH}} \longrightarrow$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{HBF}_4} \text{G} \xrightarrow{\Delta} \text{H}$ $\text{ArN}_2^+\text{X}^- \xrightarrow{\text{HBF}_4} \text{I} \xrightarrow{\text{NaNO} + \text{Cu} + \Delta} \text{J}$

NAME REACTIONS

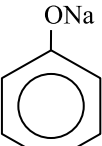
HYDROCARBON

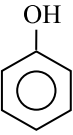
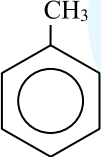
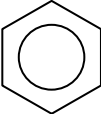
1.	Aromatisation/Reforming	$C_6H_{14} \xrightarrow[773K \& 10-20atm]{Cr_2O_3 \text{ or } V_2O_5 \text{ or } Mo_2O_3} \rightarrow$ $C_7H_{16} \xrightarrow[773K \& 10-20atm]{Cr_2O_3 \text{ or } V_2O_5 \text{ or } Mo_2O_3} \rightarrow$
2.	Ozonolysis	$CH_3CH=CH_2 + O_3 \longrightarrow A \xrightarrow{Zn+H_2O} B$ $(CH_3)_2C=CH_2 + O_3 \longrightarrow A \xrightarrow{Zn+H_2O} B$
3.	Kuchrov reaction:	$HC \equiv CH \xrightarrow{Hg^{+2}/dil.H_2SO_4} \rightarrow$ $CH_3C \equiv CH \xrightarrow{Hg^{+2}/dil.H_2SO_4} \rightarrow$
4.	Kolbe's process:	$2RCOONa \xrightarrow{\text{Electrolysis}} R-R + 2NaOH + H_2 \uparrow + 2CO_2 \uparrow$ $2CH_3COONa \xrightarrow{\text{Electrolysis}} \rightarrow$ $R-CH-COONa \xrightarrow{\text{Electrolysis}} \rightarrow$ $R-\underset{ }{CH}-COONa \xrightarrow{\text{Electrolysis}} \rightarrow$ $R-C-COONa \xrightarrow{\text{Electrolysis}} \rightarrow$ $R-\underset{ }{C}-COONa$
5.	Sabatie Senderen's reduction:	$R-CH=CH-R + H_2 \xrightarrow[200-300^\circ C]{Ni \text{ Powder}} \rightarrow$
6.	Friedal- Craft's reaction	 + R-Cl $\xrightarrow[\text{alkylation}]{AlCl_3}$  + CH ₃ -C(=O)-Cl $\xrightarrow[\text{acetylation}]{AlCl_3}$
7.	Birch reduction:	$R-C \equiv C-R + H_2 \xrightarrow{Na/liq.NH_3} \begin{matrix} R & & H \\ & \diagdown & / \\ & C=C & \\ & / & \diagdown \\ H & & R \end{matrix} \text{ (trans)}$  $\xrightarrow{Na/liq.NH_3}$ 

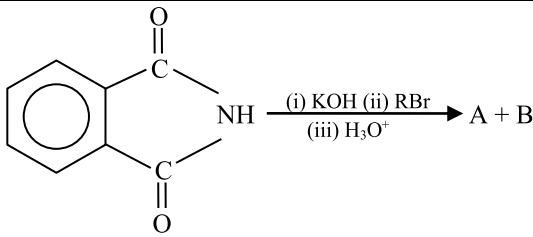
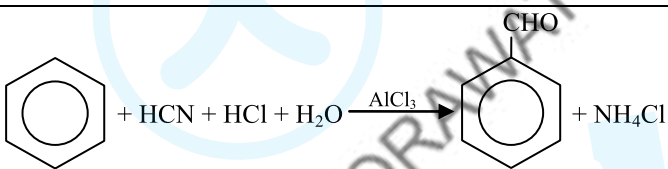
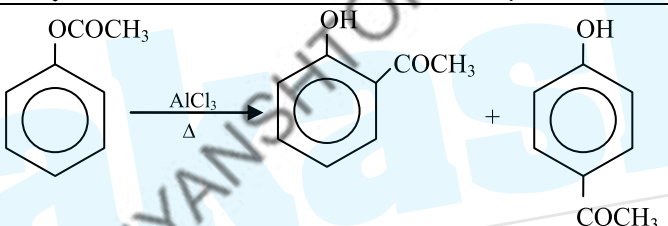
HALOALKANES & HALOARENES

8.	Darzen process:	$ROH + SOCl_2 \xrightarrow{\text{Pyridine}} \rightarrow$
9.	Finkelstein reaction:	$R-X + NaI \xrightarrow{\text{acetone}} \rightarrow$
10.	Swart reaction:	$R-X + AgF \xrightarrow{DMSO} \rightarrow$
11.	Wurtz reaction:	$2R-X + 2Na \xrightarrow{\text{dry Ether}} \rightarrow$
12.	Wurtz-Fittig reaction:	 -Cl + 2Na + Cl-R $\xrightarrow[\Delta]{\text{ether}} \rightarrow$
13.	Fittig reaction:	$2 \text{ (Benzene ring) } -Cl + 2Na \xrightarrow[\Delta]{\text{ether}} \rightarrow$

ALCOHAL, PHENOL & ETHER

14.	Kolbe -Schmidt reaction:	 + CO ₂ $\xrightarrow[H^+]{140^\circ C} \rightarrow$
-----	--------------------------	--

15.	Reimer-Tiemann's reaction:	 $\xrightarrow[\Delta]{\text{CHCl}_3 + 3\text{KOH}}$ $\xrightarrow[\Delta]{\text{CCl}_4 + 4\text{KOH}}$
16.	Williamson's synthesis:	$\text{RX} + \text{R}'\text{ONa} \xrightarrow{140^\circ\text{C}} \text{R}-\text{O}-\text{R}' + \text{NaX}$ $\text{CH}_3\text{Br} \xrightarrow{(\text{CH}_3)_3\text{CONa}}$ $(\text{CH}_3)_3\text{CBr} \xrightarrow{\text{CH}_3\text{ONa}}$ $\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{NaOH}} \text{A} \xrightarrow{\text{CH}_3-\text{X}} \text{B}$
ALDEHYDE, KETONE AND CARBOXYLIC ACID		
17.	Rosenmund reduction:	$\text{R}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl} + \text{H}_2 \xrightarrow{\text{Pd}+\text{BaSO}_4}$ $\text{Ph}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl} + \text{H}_2 \xrightarrow{\text{Pd}+\text{BaSO}_4}$
18.	Stephen's reduction:	$\text{R}-\text{C}\equiv\text{N} + \text{H}_2 \xrightarrow{\text{SnCl}_2 + \text{HCl}} \xrightarrow{\text{H}_3\text{O}^+}$
19.	Etard reaction:	 $\xrightarrow[\text{[CS}_2\text{]}]{\text{CrO}_2\text{Cl}_2} \text{A} \xrightarrow{\text{H}_3\text{O}^+} \text{B}$
20.	Gattermann-Koch's reaction:	 $+ \text{CO} + \text{HCl} \xrightarrow[\text{or CuCl}]{\text{AlCl}_3}$
21.	Clemmensen's reduction:	$>\text{C}=\text{O} \xrightarrow{\text{Zn}-\text{Hg}/\text{HCl}}$
22.	Wolff-kishner's reduction:	$>\text{C}=\text{O} \xrightarrow{\text{NH}_2-\text{NH}_2/\text{OH}^-/\text{Ethylene Glycol}}$
23.	Haloform Test:	$\text{CH}_3-\underset{\underset{\text{OH}}{\mid}}{\text{CH}}-\text{Z} \xrightarrow{\text{X}_2+\text{NaOH}}$ $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Z} \xrightarrow{\text{X}_2+\text{NaOH}}$ Reagent- $\text{X}_2 + \text{NaOH}$ or NaOX or $\text{X}_2 + \text{Na}_2\text{CO}_3$ or $\text{CaOCl}_2 + \text{H}_2\text{O}$
24.	Aldol Condensation	$\text{CH}_3-\underset{\underset{\text{O}}{\parallel}}{\text{CH}} \xrightarrow[\text{Aldol condensation}]{\text{Dil. NaOH}}$ $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3 \xrightarrow[\text{Aldol condensation}]{\text{Dil. NaOH}}$ $\xrightarrow[\Delta]{-\text{H}_2\text{O}}$ $\xrightarrow[\Delta]{-\text{H}_2\text{O}}$ $\text{CH}_3\text{CHO} + \text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{(ii) } \Delta]{\text{(i) Ba(OH)}_2} \text{A} + \text{B} + \text{C} + \text{D}$ $\text{PhCHO} + \text{PhCOCH}_3 \xrightarrow[293\text{K}]{\text{OH}^-} \text{A}$
25.	Cannizzaro reaction:	$2\text{HCHO} \xrightarrow{50\%\text{NaOH}} \text{A} + \text{B}$ $\text{PhCHO} \xrightarrow{50\%\text{NaOH}} \text{A} + \text{B}$ $(\text{p})\text{CH}_3\text{O}-\text{Ph}-\text{CHO} + \text{HCHO} \xrightarrow{50\%\text{NaOH}} \text{A} + \text{B}$

26.	Hell- Volhard-Zelinsky (HVZ) reaction:	$\text{CH}_3\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) Br}_2 + \text{Red P}}$
AMINE		
27.	Gabriel's phthalimide Synthesis:	 Aromatic primary amine cannot be prepared
28.	Hoffmann's bromamide reaction:	$\text{R}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{NH}_2 \xrightarrow{\text{Br}_2 + 4\text{KOH}}$ Preparation of primary Amine
29.	Isocyanide Test: or Carbyl amine test:	$\text{R}-\text{NH}_2 \xrightarrow[\Delta]{\text{CHCl}_3/\text{KOH}}$ (Offensive smell) not $\text{Ph}-\text{NH}_2 \xrightarrow[\Delta]{\text{CHCl}_3/\text{KOH}}$ (Offensive smell) not
Some Extramportant Reaction		
30.	Corey- House process:	$\text{R}-\text{X} \xrightarrow[\text{(iii) R'-X}]{\text{(i) Li (ii) CuX}}$ $\rightarrow \text{R}-\text{R}'$
31.	Gattermann's aldehyde synthesis:	
32.	Fries rearrangement:	
33.	Wacker process:	$\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} + \text{PdCl}_2 \xrightarrow{\text{Cu}_2\text{Cl}_2} \text{CH}_3\text{CHO} + \text{Pd} + 2\text{HCl}$ $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{H}_2\text{O} + \text{PdCl}_2 \xrightarrow{\text{Cu}_2\text{Cl}_2} \text{CH}_3\text{COCH}_3 + \text{Pd} + 2\text{HCl}$



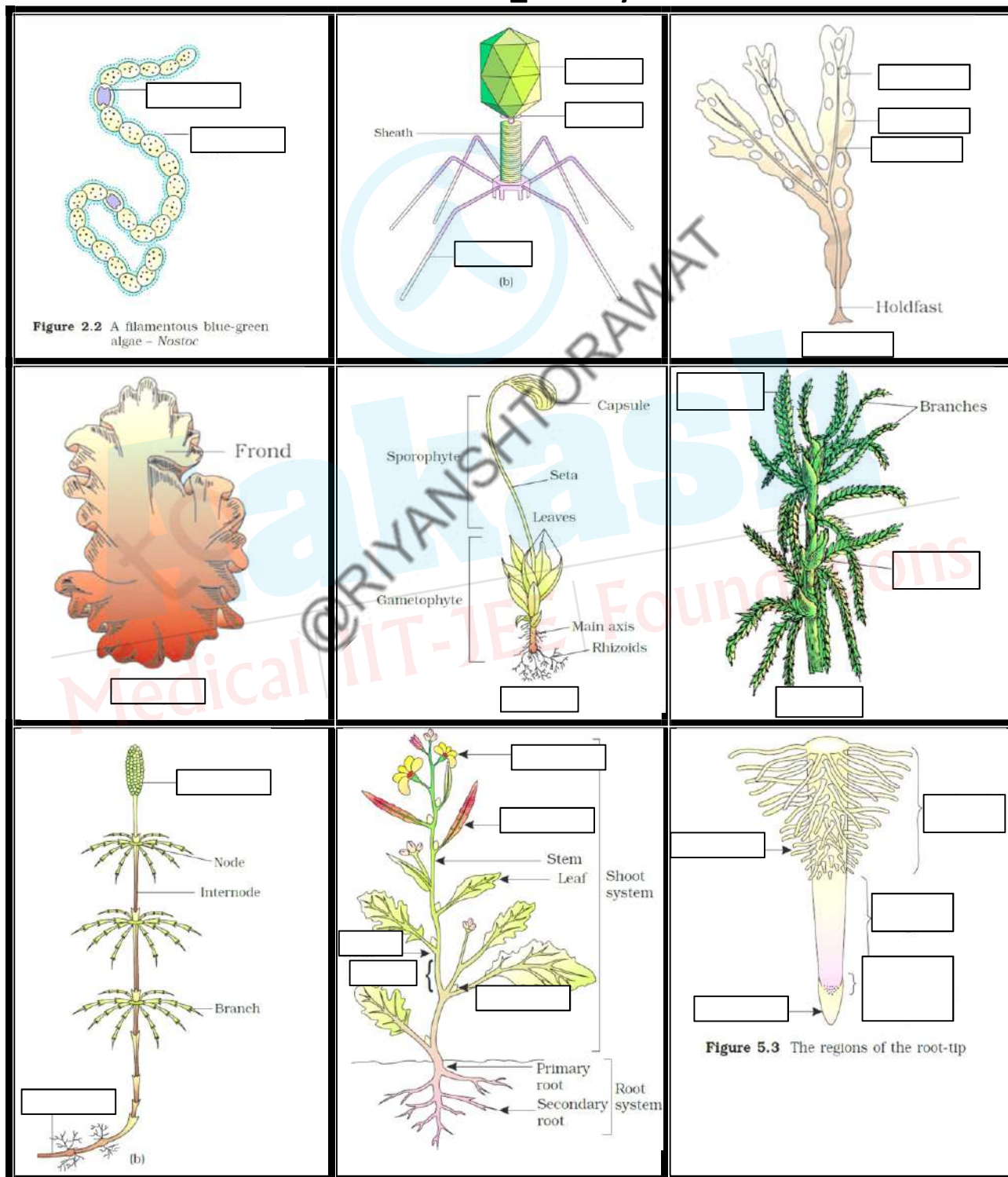
Aakash

Medical | IIT-JEE | Foundations

Regd. Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

Diwali Home Work

Class –XI_ Botany



[CLICK HERE TO GET AAKASH TEST PAPERS](#)

KOTA



Types of Aestivation (a) (b) (c)

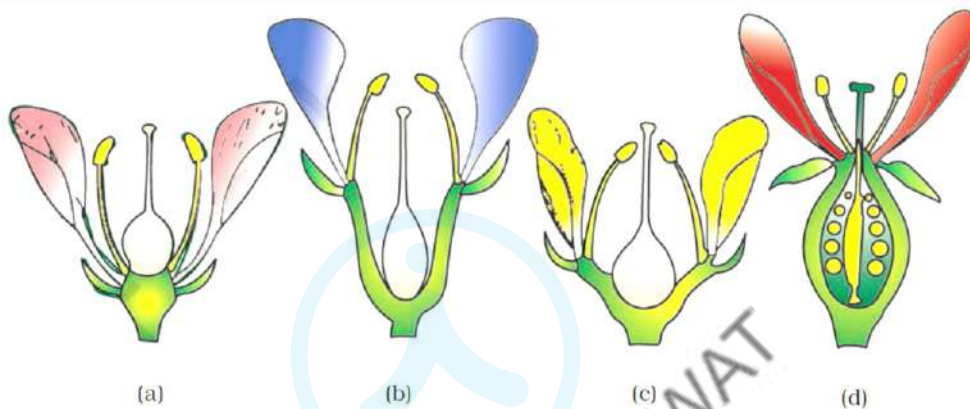
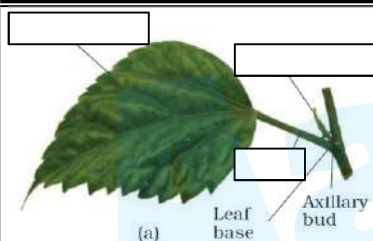


Figure 5.9 Position of floral parts on thalamus : (a) (b) and (c) Perigynous (d)



(a)

Leaf base

Axillary bud



(c)

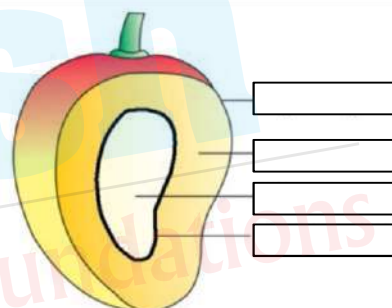


(d)

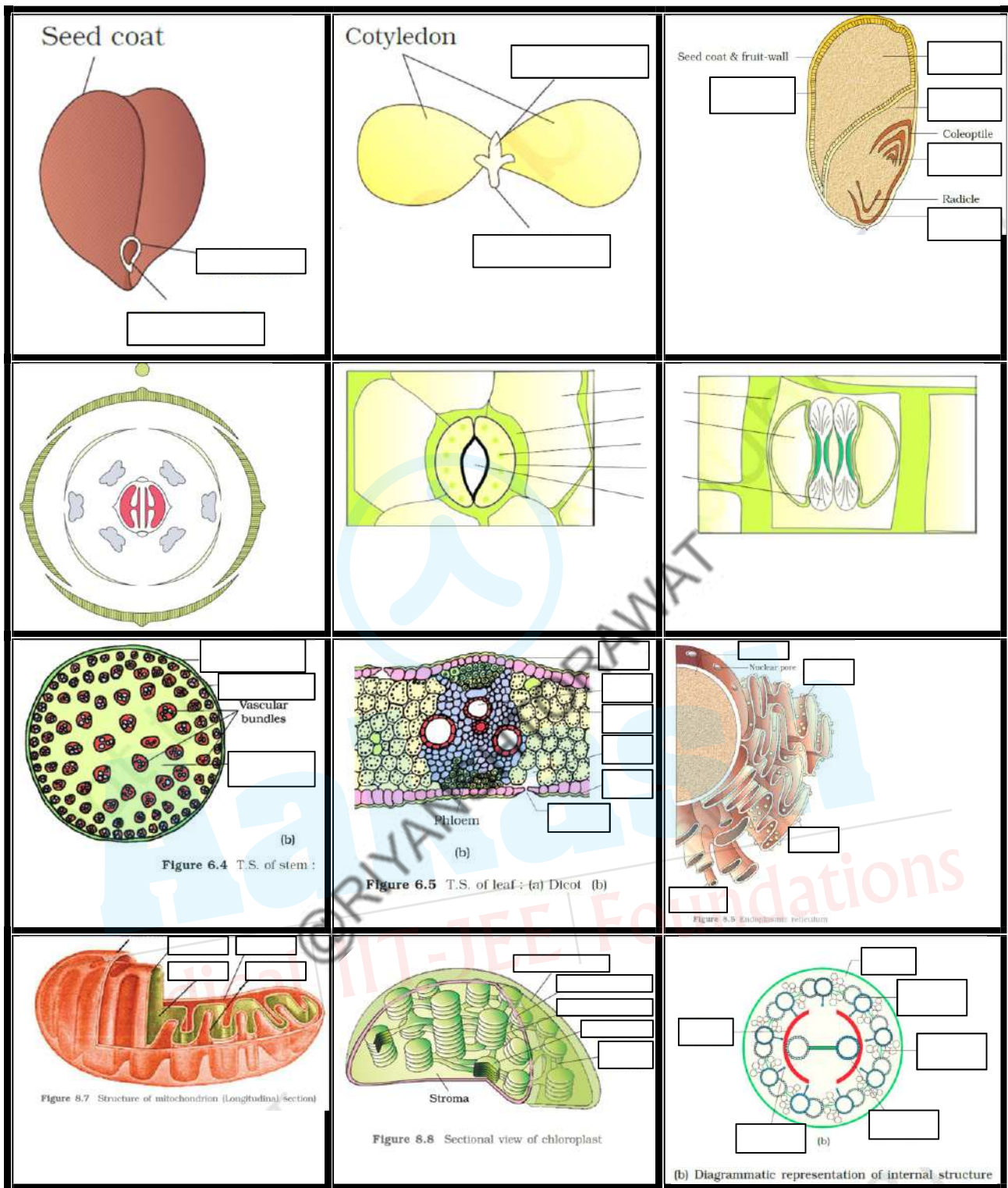


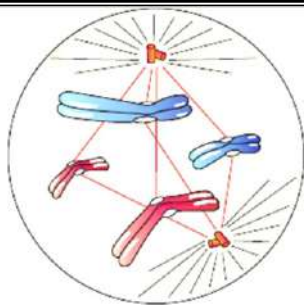
(e)

Figure 5.12 Types of placentation :

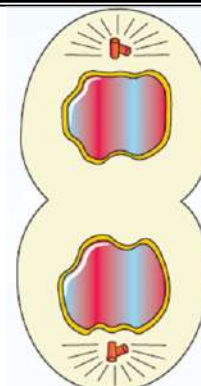


(a)





A diagrammatic view of stages in mitosis



A diagrammatic view of stages in Mitosis

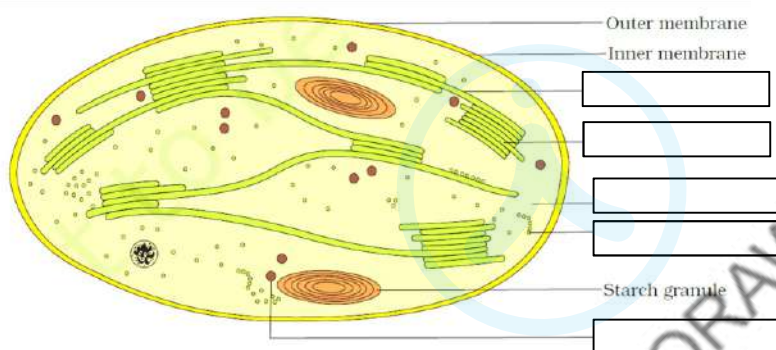
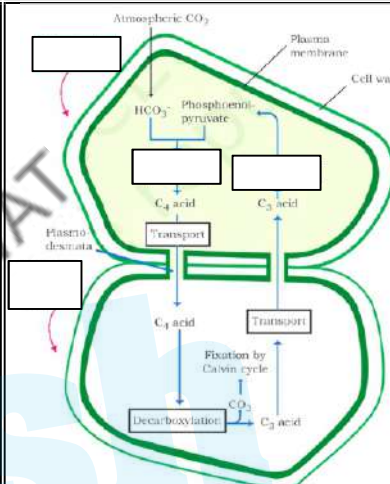


Figure 11.2 Diagrammatic representation of an electron micrograph of a section of chloroplast



Diagrammatic representation of the Hatch and Slack Pathway

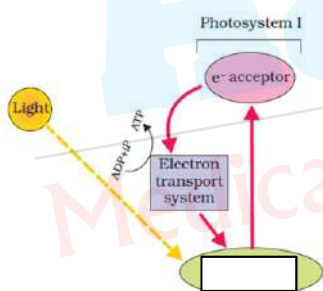


Figure 11.6 Cyclic photophosphorylation

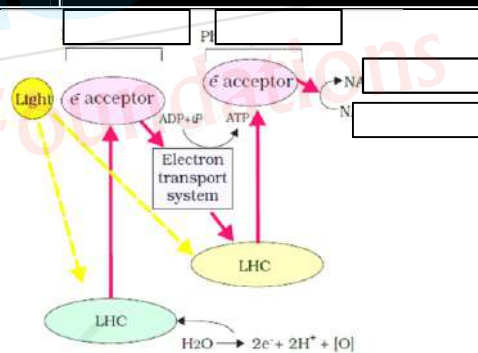


Figure 11.5 Z scheme of light reaction

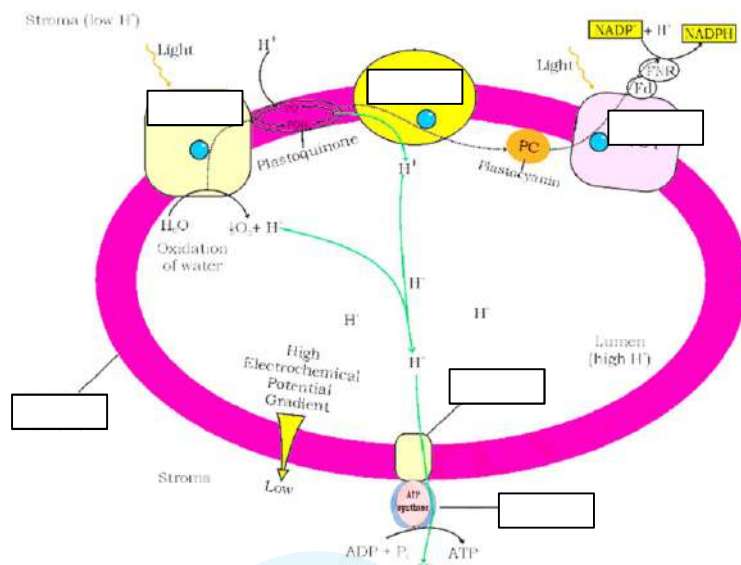


Figure 11.7 ATP synthesis through chemiosmosis

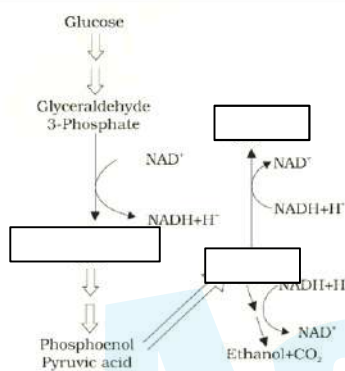


Figure 12.2 Major pathways of anaerobic respiration

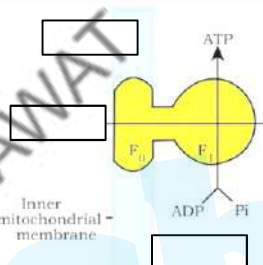


Figure 12.5 Diagrammatic presentation of ATP synthesis in mitochondria

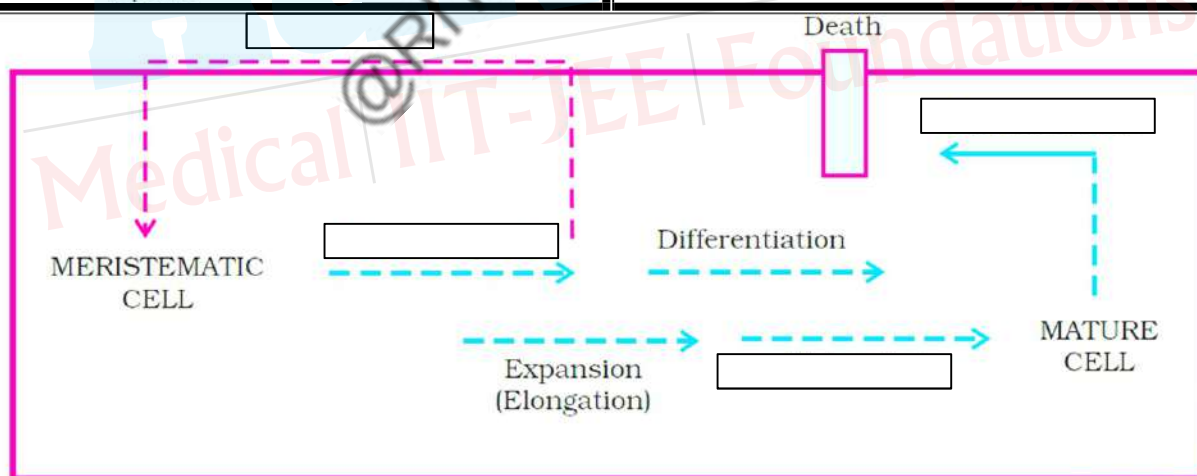


Figure 13.8 Sequence of the developmental process in a plant cell



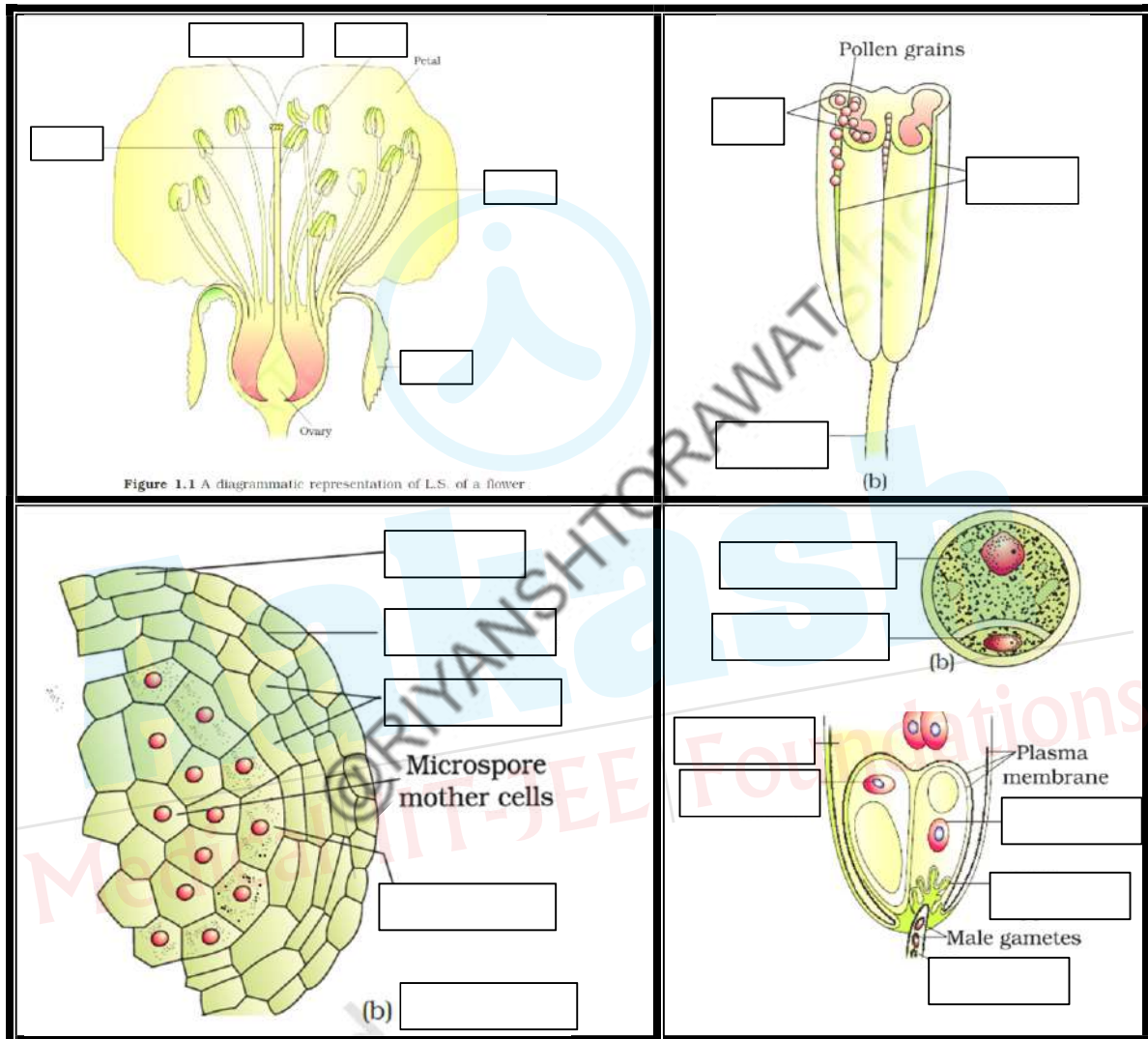
Aakash

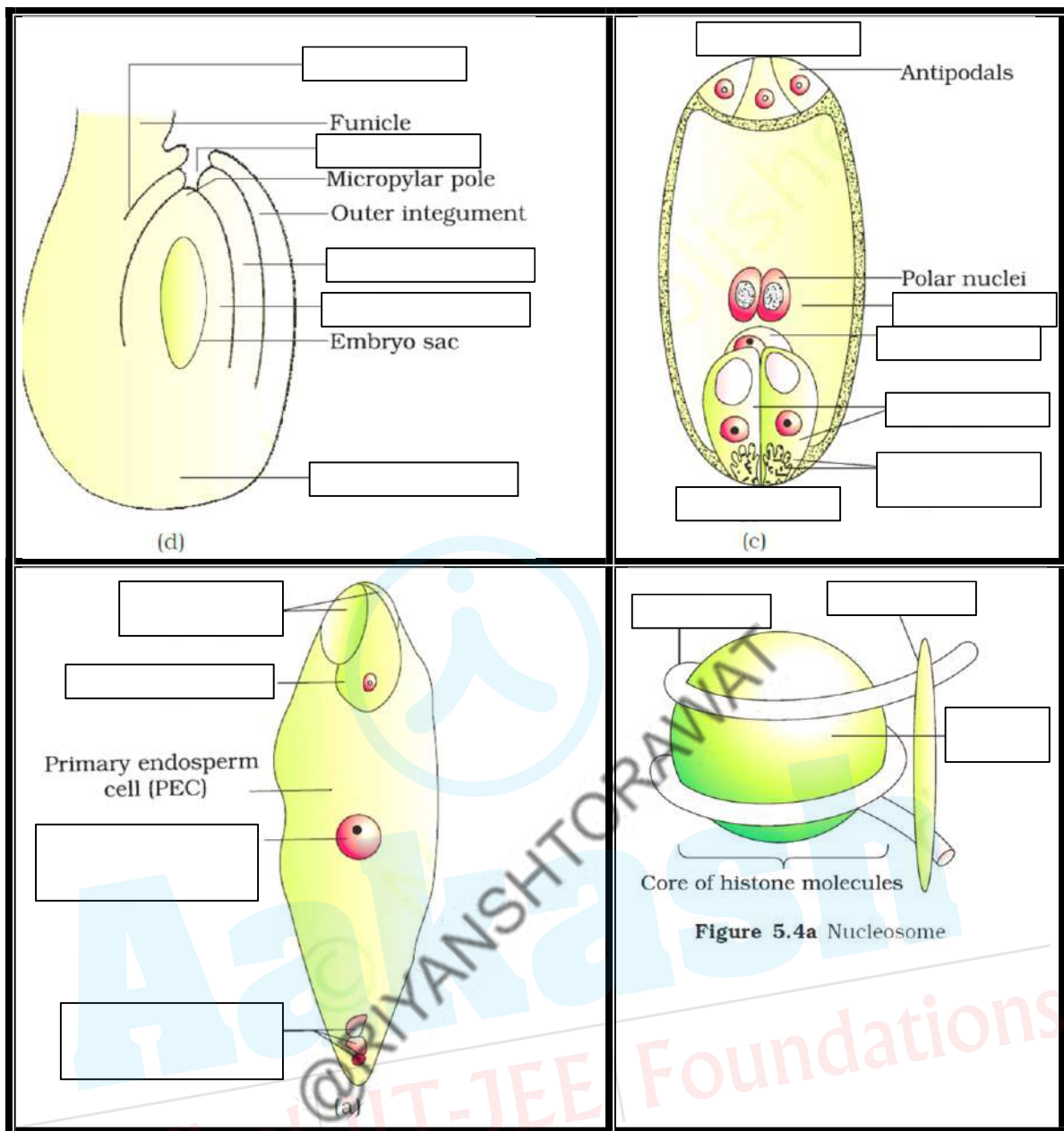
Medical | IIT-JEE | Foundations

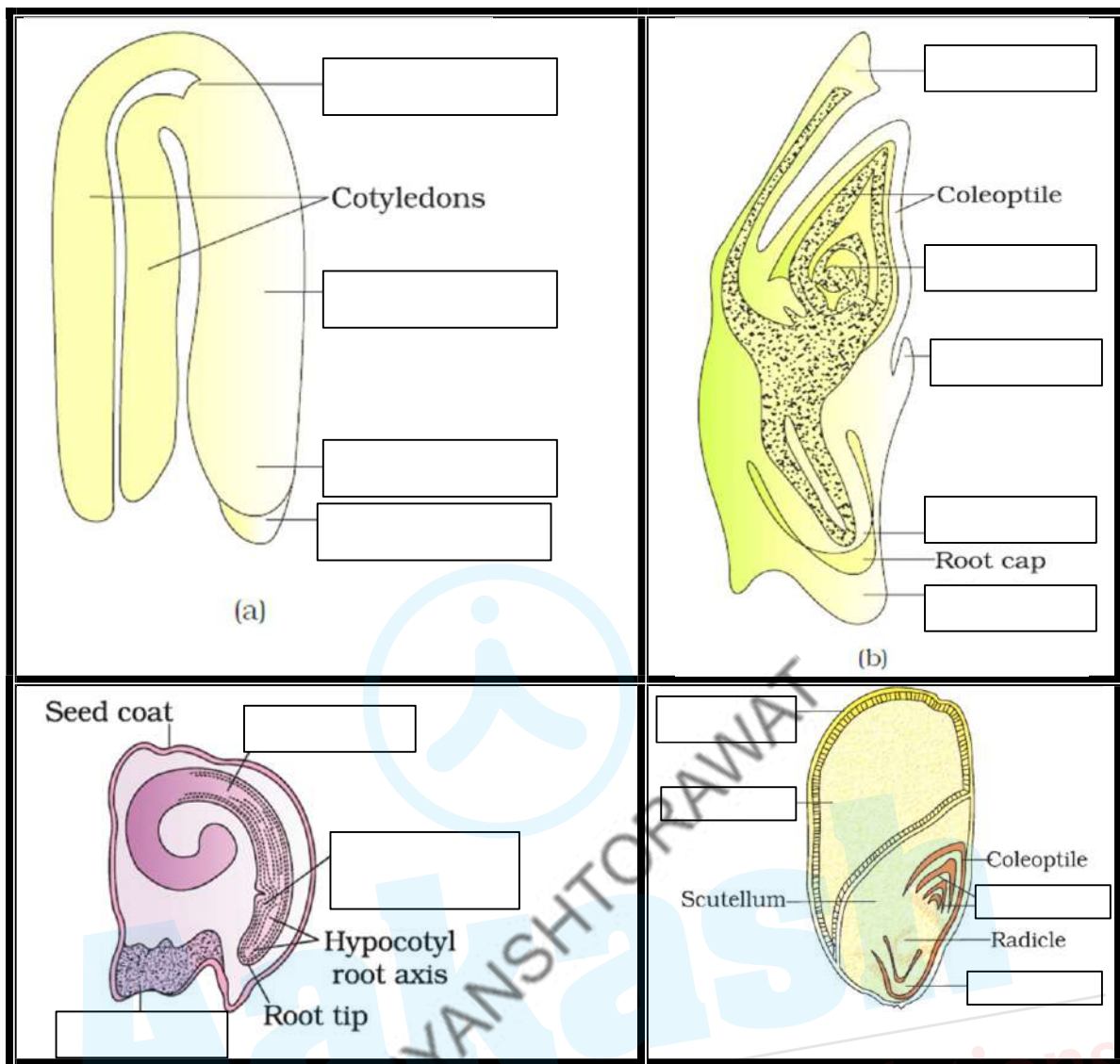
Regd. Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

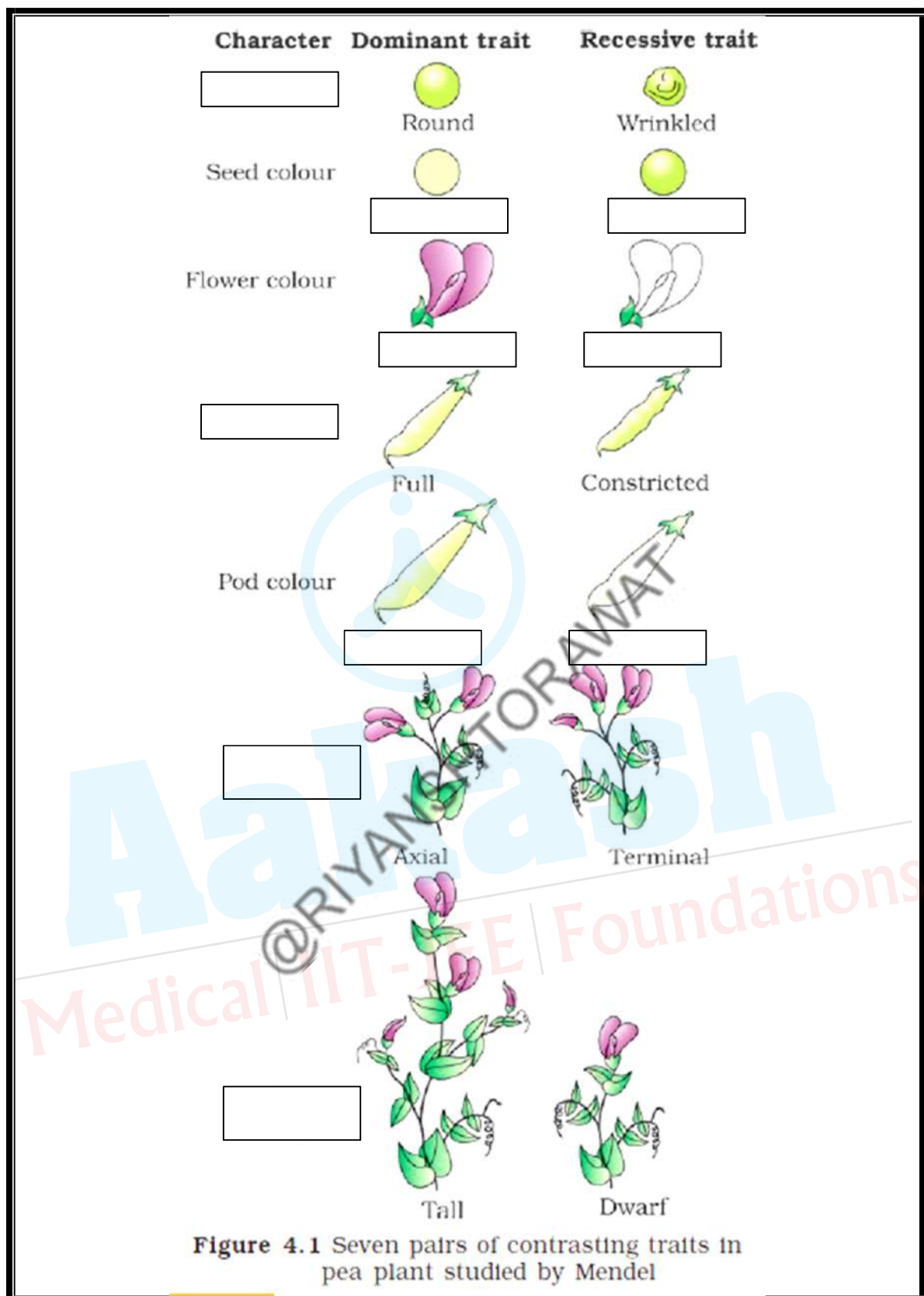
Diwali Home Work

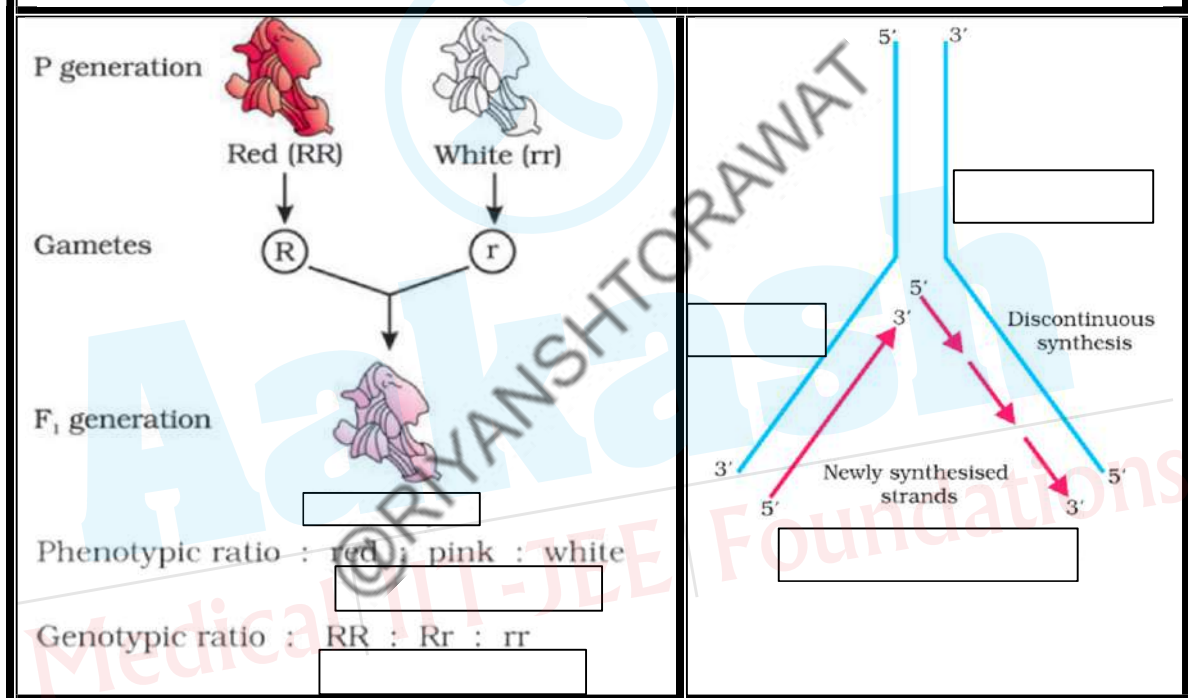
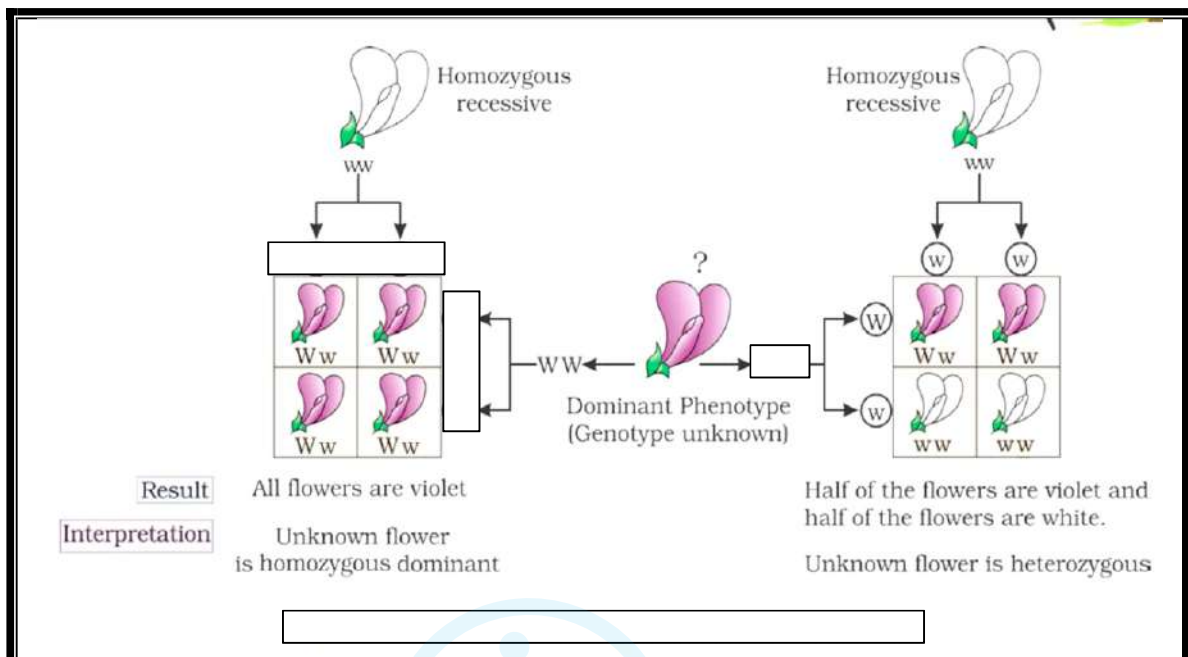
Class –XII_ Botany



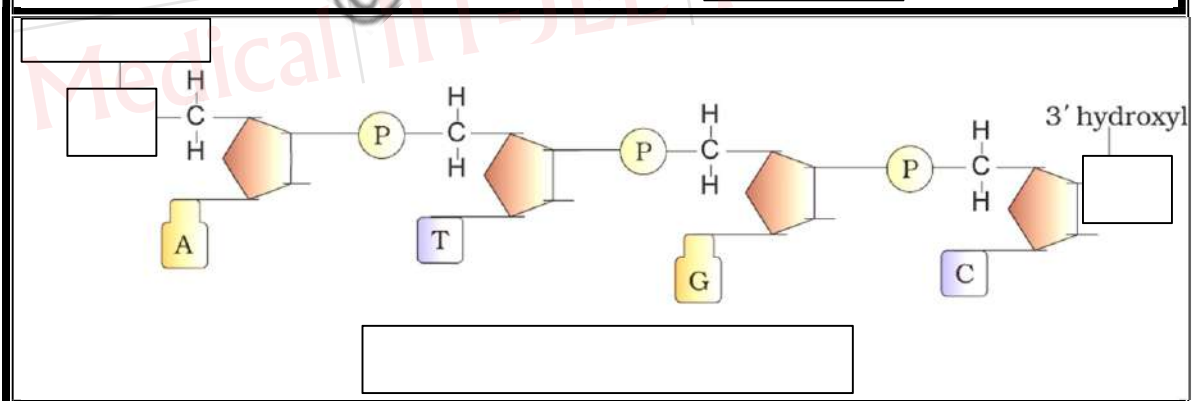
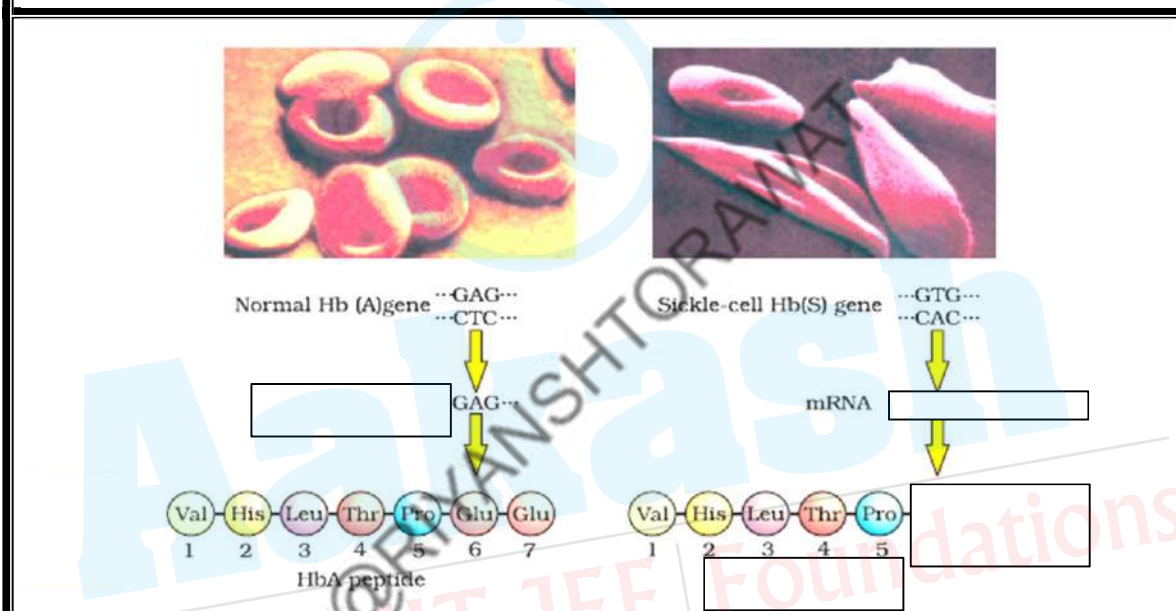








Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood types of offspring
	I^A		A
I^A		$I^A I^B$	
	i		A
I^B			AB
	I^B		B
I^B			B
i	i		



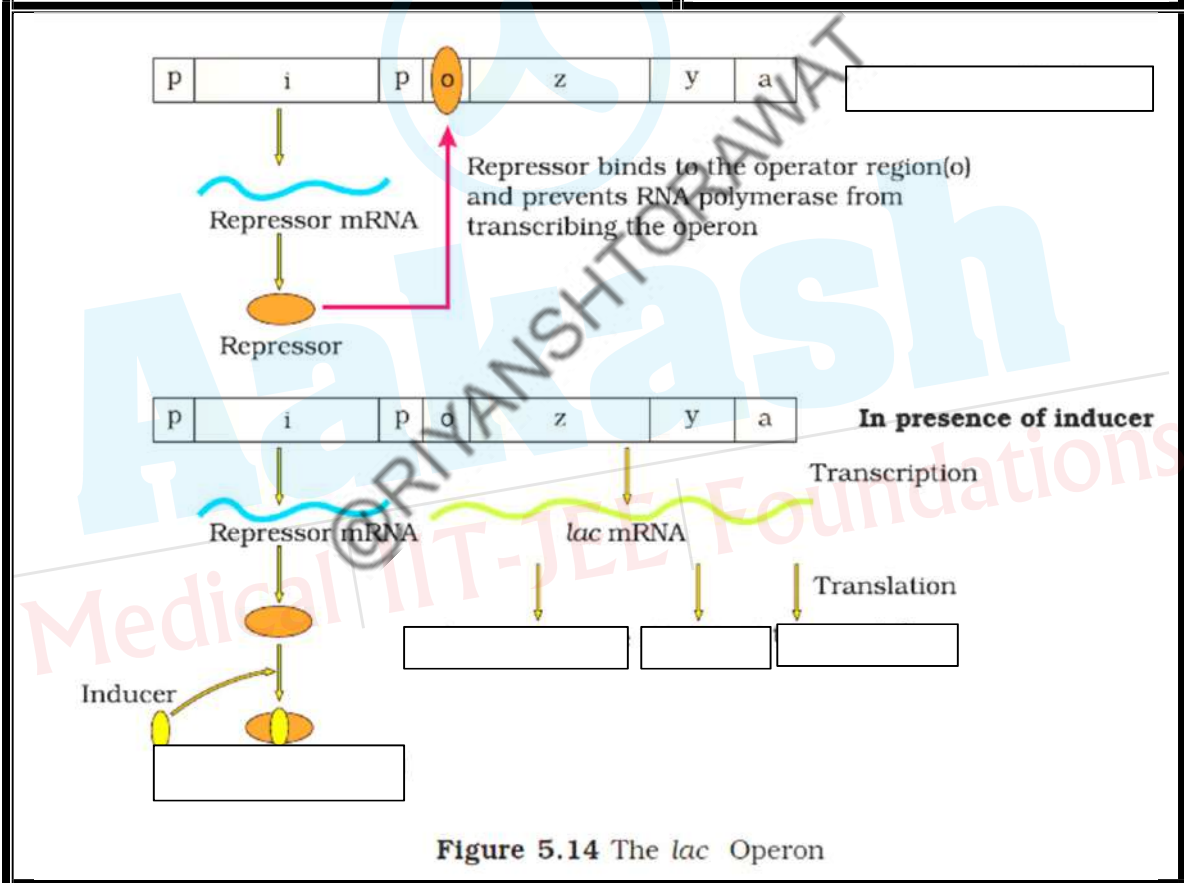
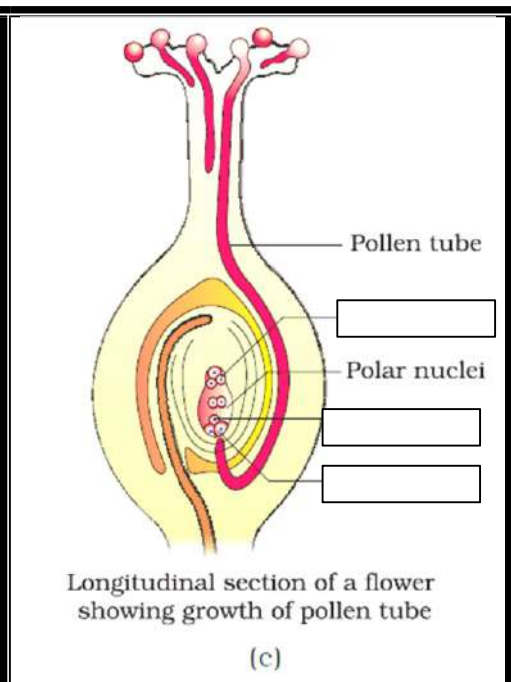
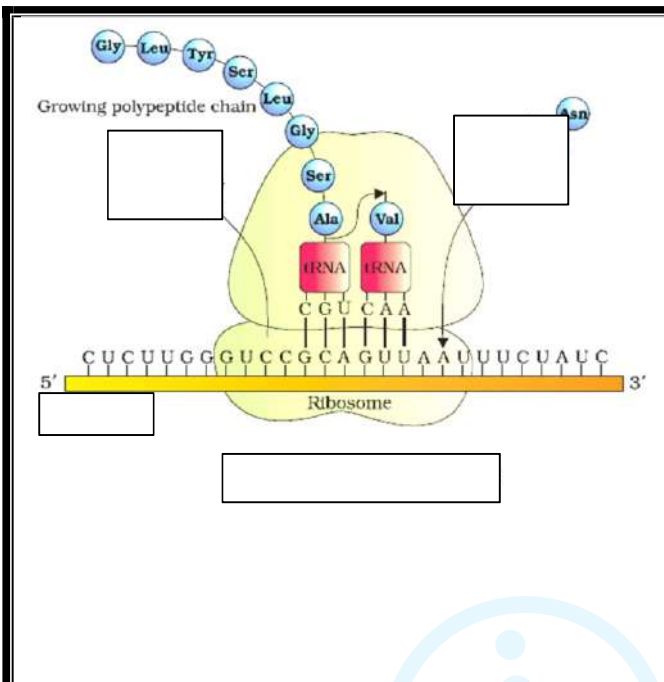


Figure 5.14 The *lac* Operon

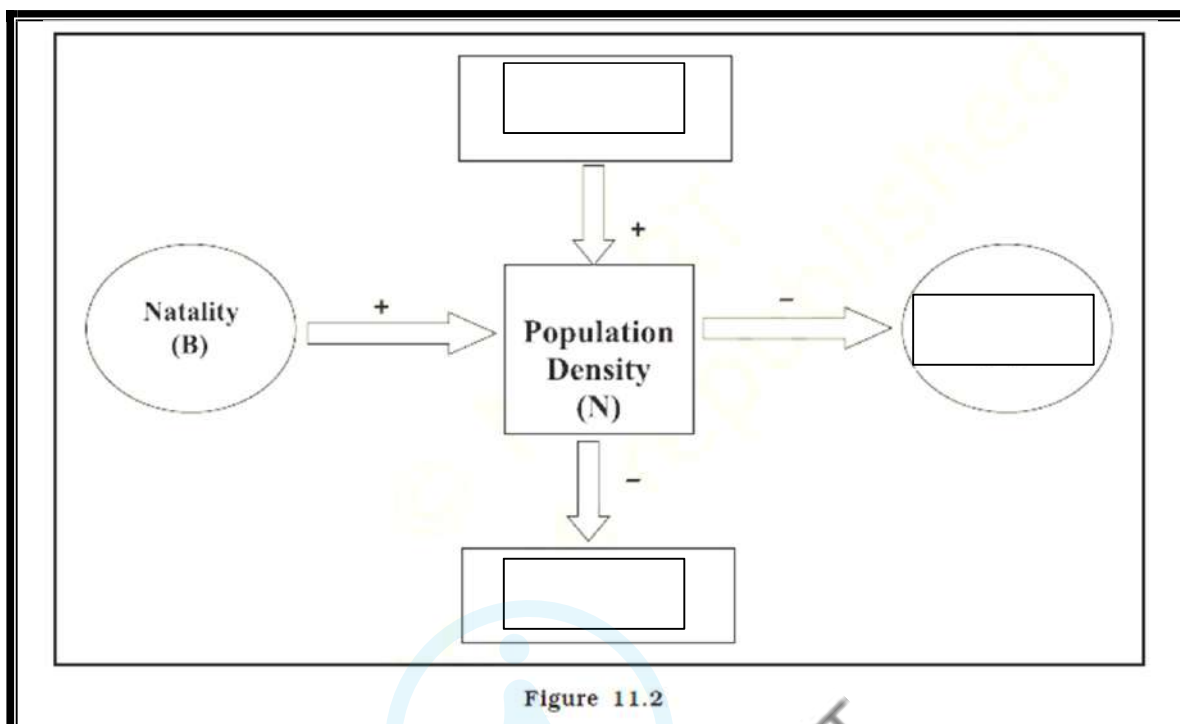
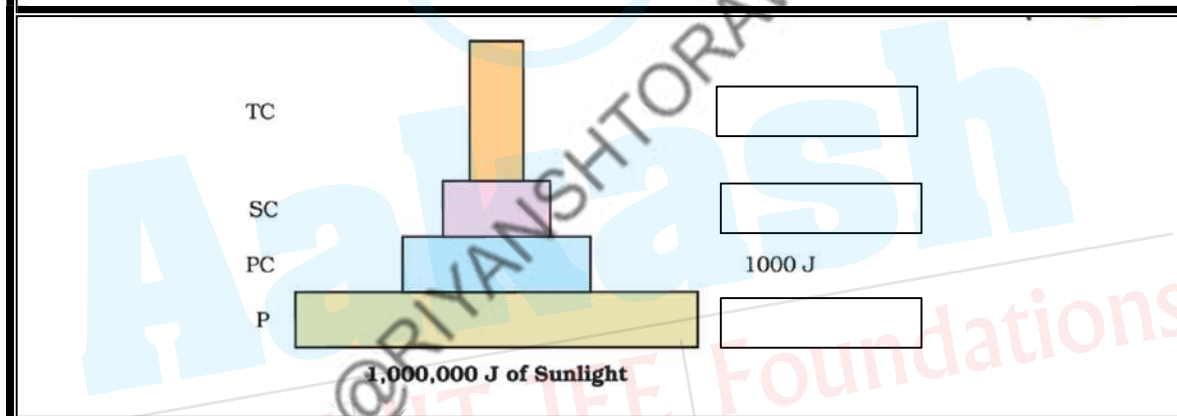
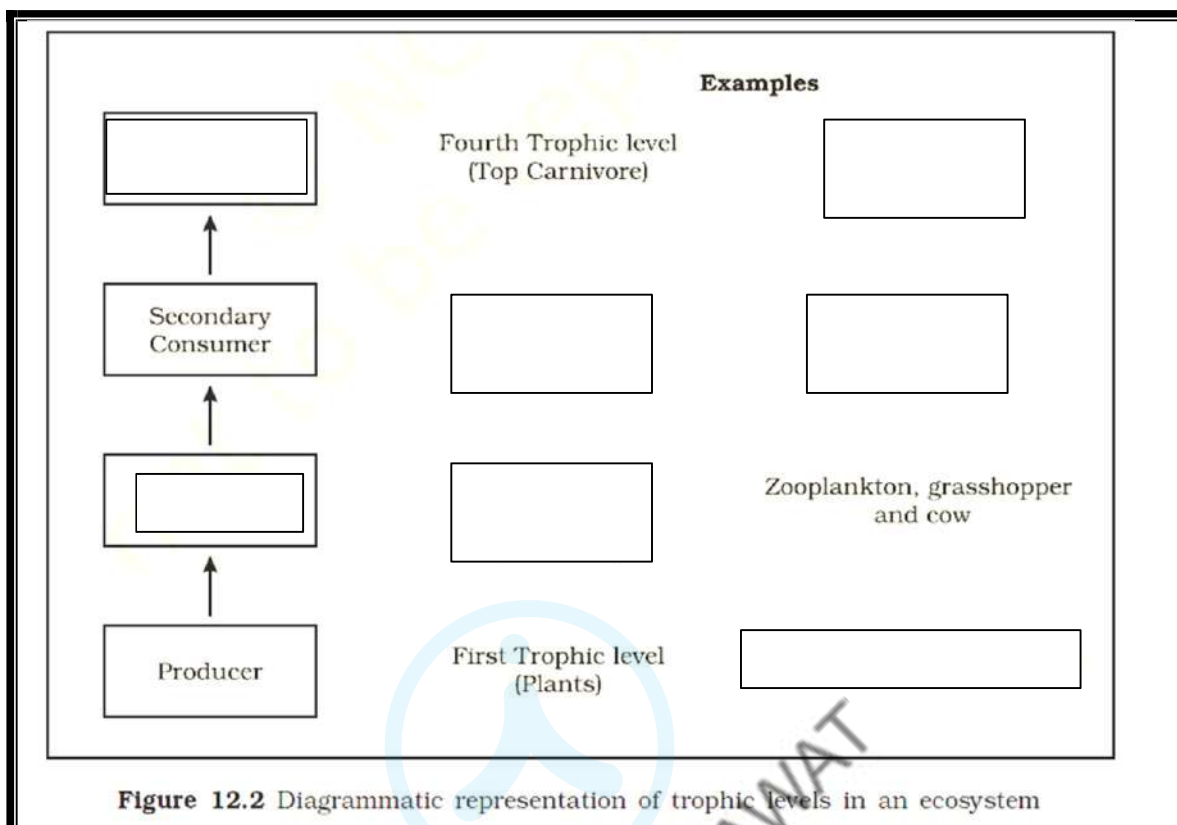


Figure 11.2

Population Interactions		
Species A	Species B	Name of Interaction
		Mutualism
-	-	
+		Predation
	-	Parasitism
+		Commensalism
		Amensalism





Aakash

Medical | IIT-JEE | Foundations

Regd. Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

Diwali Home Work

Class –XI_ Zoology

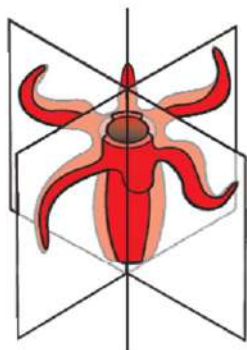


Figure 4.1 (a) symmetry

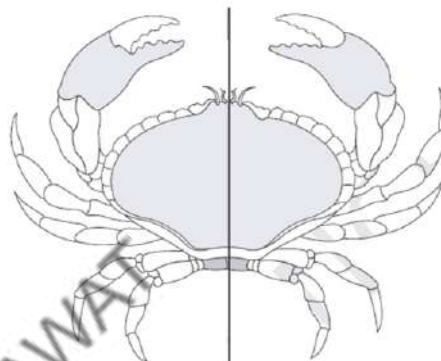


Figure 4.1 (b) symmetry

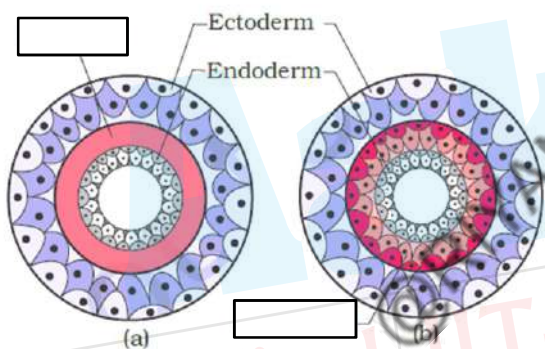


Figure 4.2 Showing germinal layers :
(a) Diploblastic (b) Triploblastic

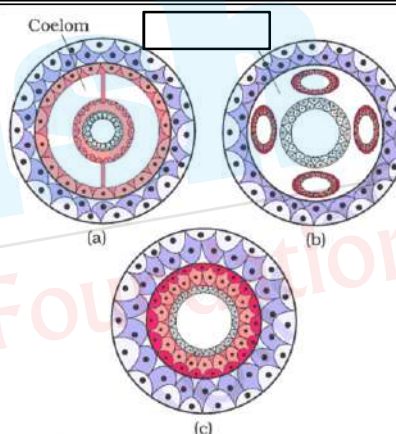


Figure 4.3 Diagrammatic sectional view of :
(a) (b) Pseudocoelomate
(c) Acoelomate

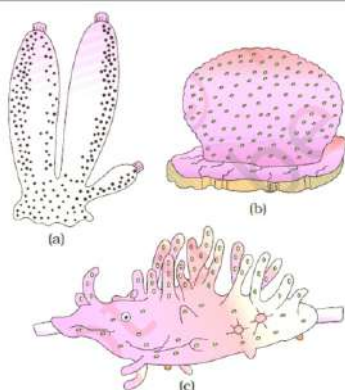
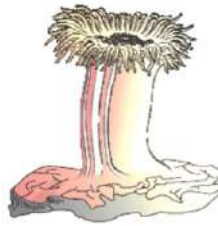


Figure 4.5 Examples of Porifera : (a)
(b) (c)



Figure 4.7
Diagrammatic view of



(a)

(b)

Figure 4.6 Examples of Coelenterata indicating outline of their body form :
(a) (b)

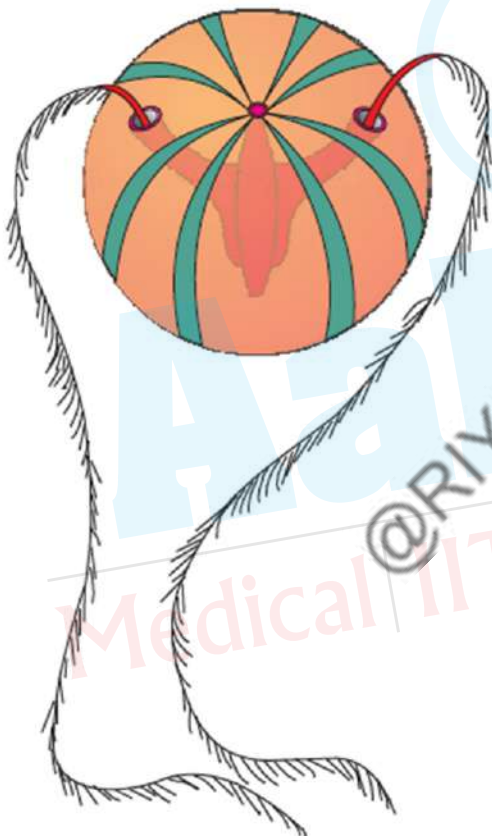


Figure 4.8 Example of



(a)



(b)

Figure 4.9 Examples of Platyhelminthes : (a) (b)

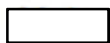
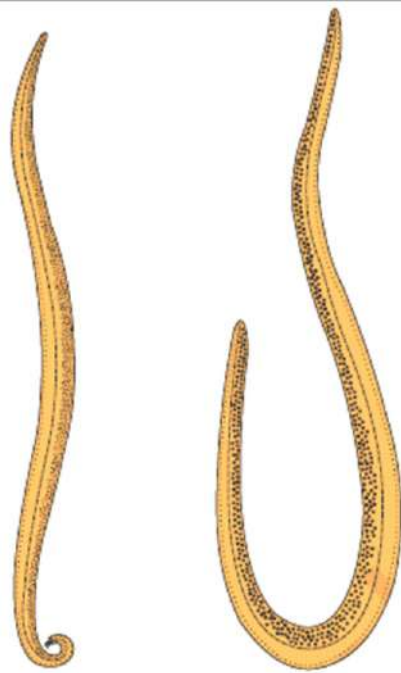


Figure 4.10 Example of Aschelminthes:
Roundworm

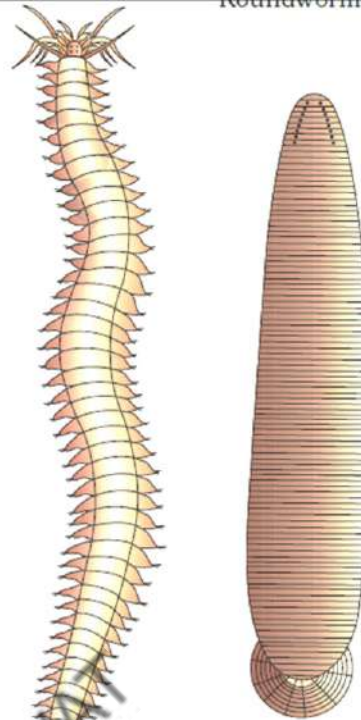
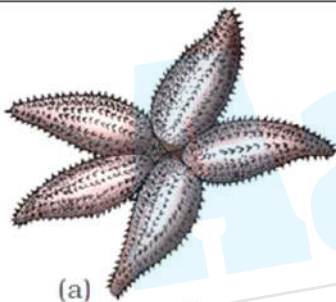
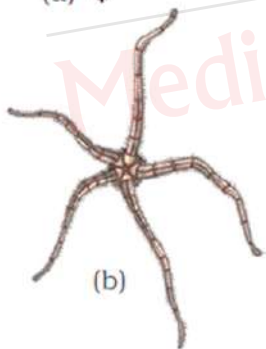


Figure 4.11 Examples of Annelida : (a)
(b)



(a)



(b)

Figure 4.14 Examples of Echinodermata :
(a)
(b)

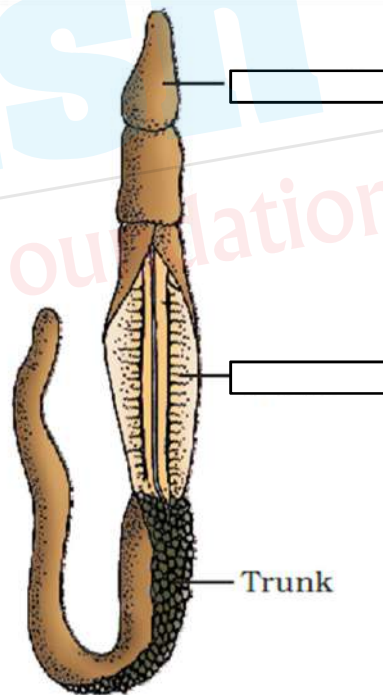


Figure 4.15 *Balanoglossus*

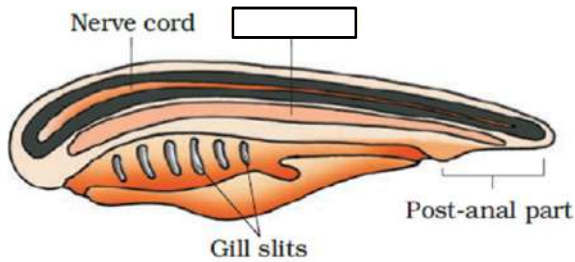


Figure 4.16 Chordata characteristics

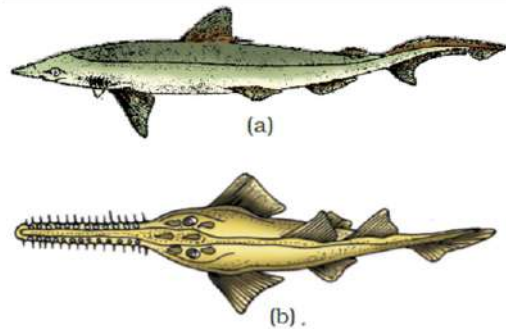


Figure 4.19 Example of Cartilaginous fishes :
(a) [] (b) []

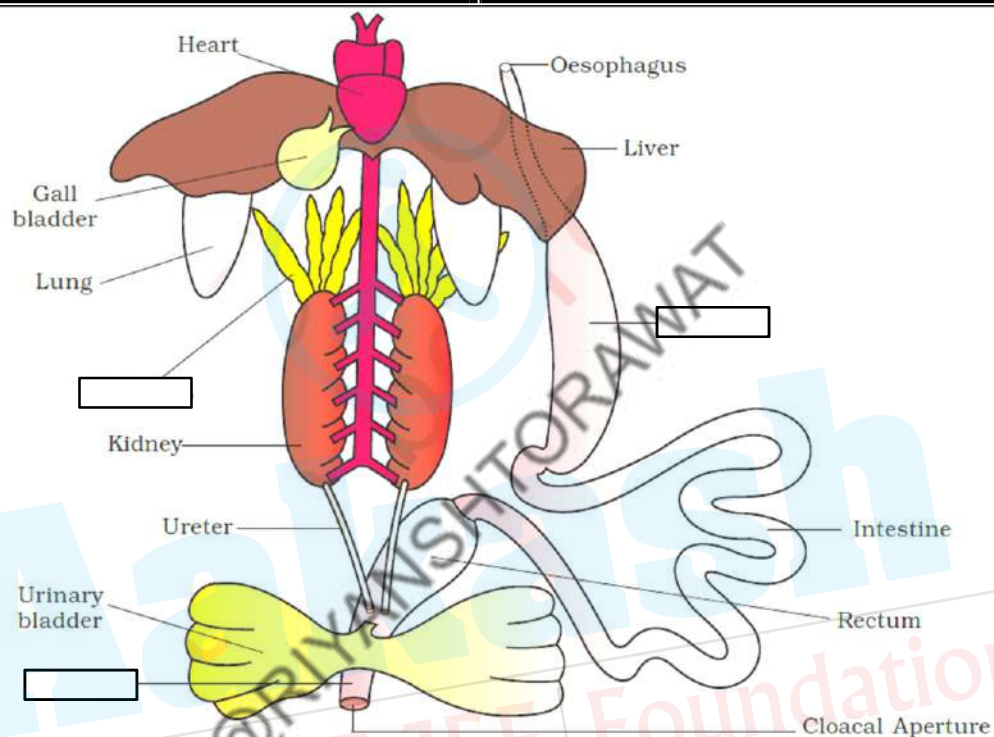


Figure 7.2 Diagrammatic representation of internal organs of frog showing complete digestive system

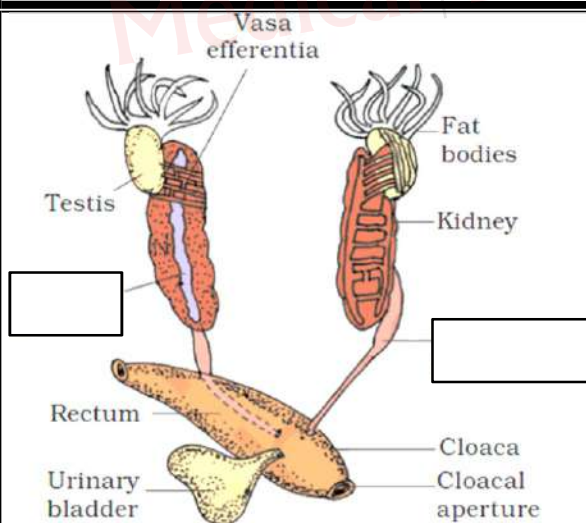


Figure 7.3 Male reproductive system

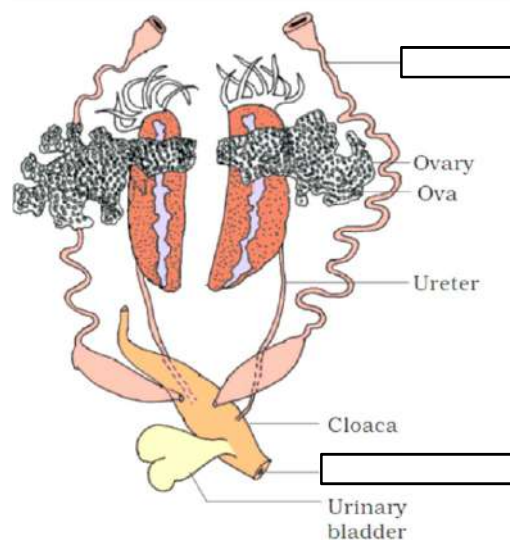
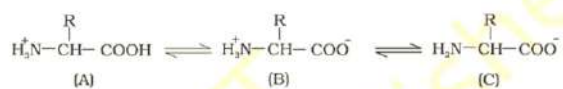
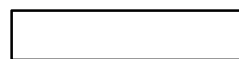
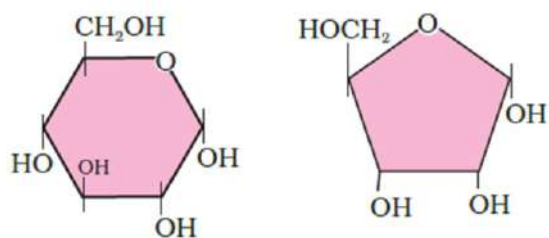


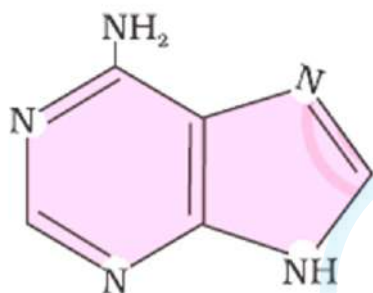
Figure 7.4 Female reproductive system



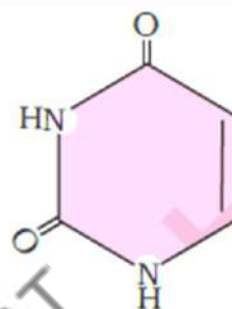
B is called



Sugars (Carbohydrates)



(Purine)



(Pyrimidine)

Nitrogen bases

TABLE 9.3 Some Secondary Metabolites

Pigments	Carotenoids,
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Ricin
Lectins	Concanavalin A
Drugs	curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

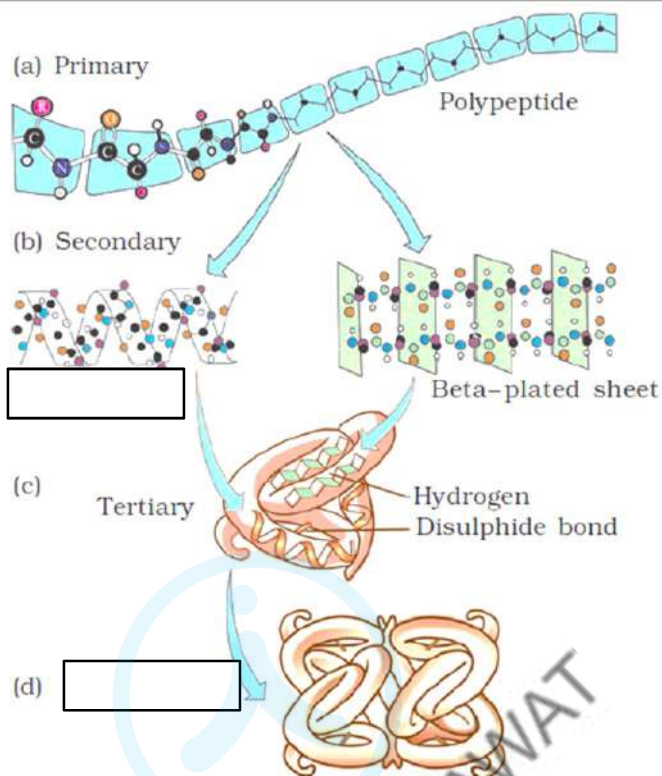


Figure 9.3 Various levels of Protein Structure

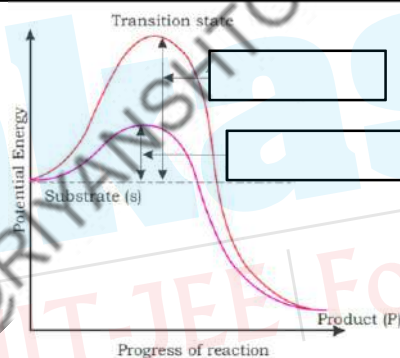


Figure 9.4 Concept of activation energy

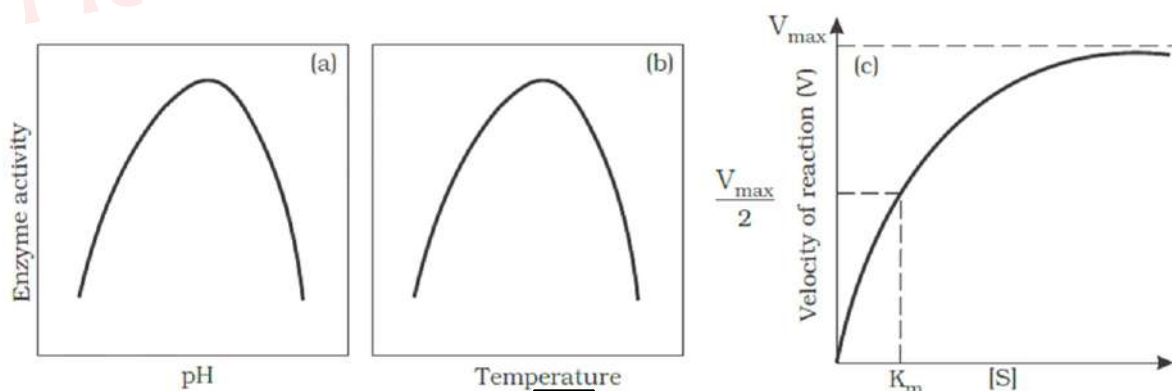


Figure 9.5 Effect of change in : (a) (b) and (c) Concentration of substrate on enzyme activity

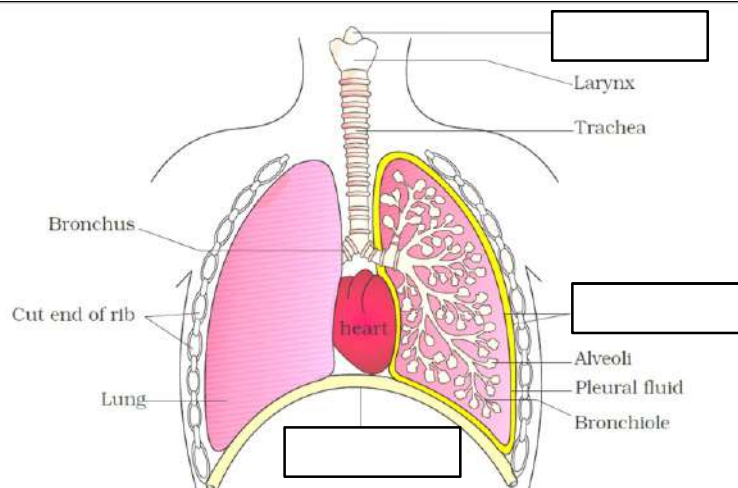


Figure 14.1 Diagrammatic view of human respiratory system (sectional view of the left lung is also shown)

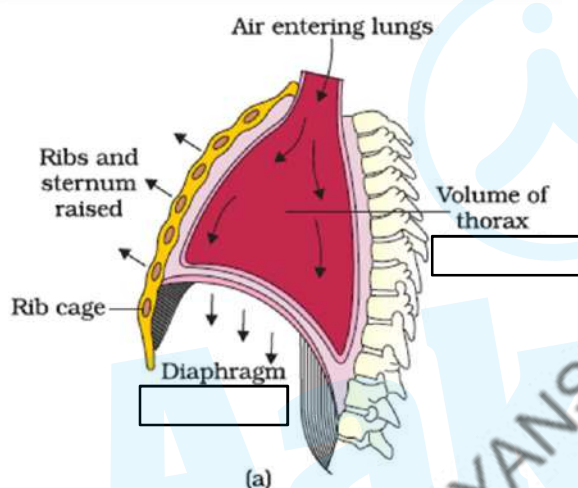


Figure 14.2 Mechanism of breathing showing :
(a) inspiration

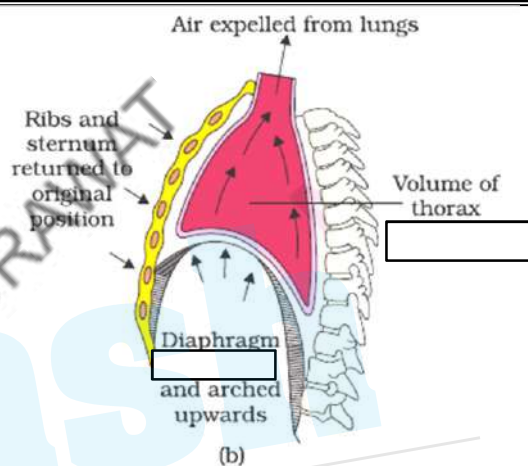


Figure 14.2 Mechanism of breathing showing :
(b) expiration

TABLE 14.1 Partial Pressures (in mm Hg) of Oxygen and Carbon dioxide at Different Parts Involved in Diffusion in Comparison to those in Atmosphere

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O ₂	159		40	95	40
CO ₂	0.3	40	45	40	

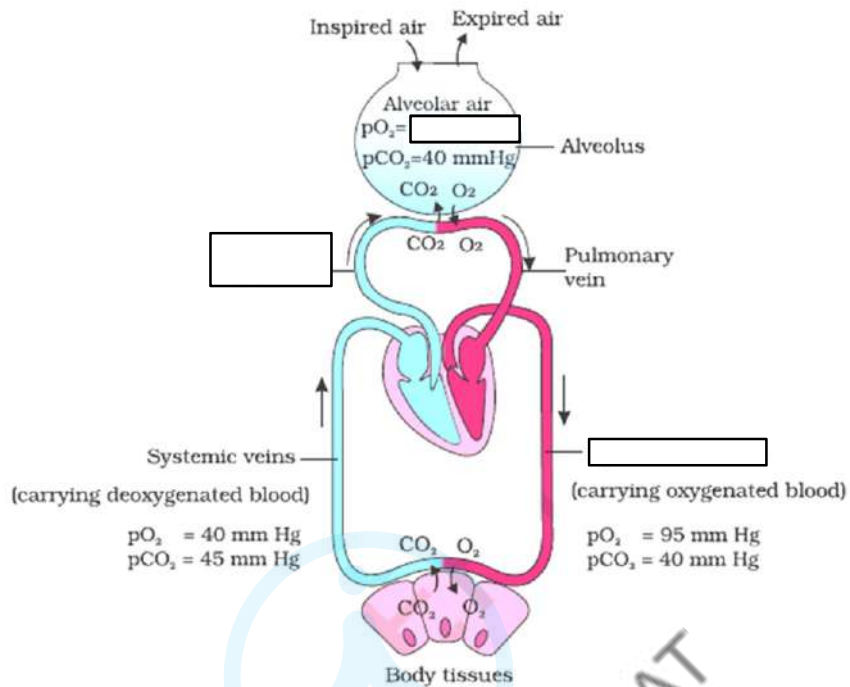


Figure 14.3 Diagrammatic representation of exchange of gases at the alveolus and the body tissues with blood and transport of oxygen and carbon dioxide

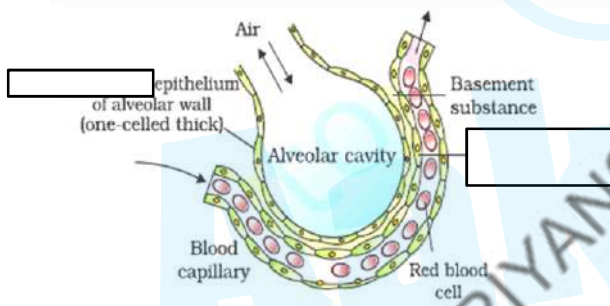


Figure 14.4 A Diagram of a section of an alveolus with a pulmonary capillary.

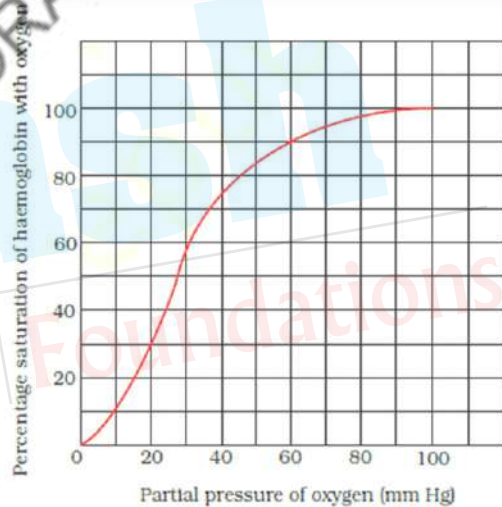


Figure 14.5 [] curve

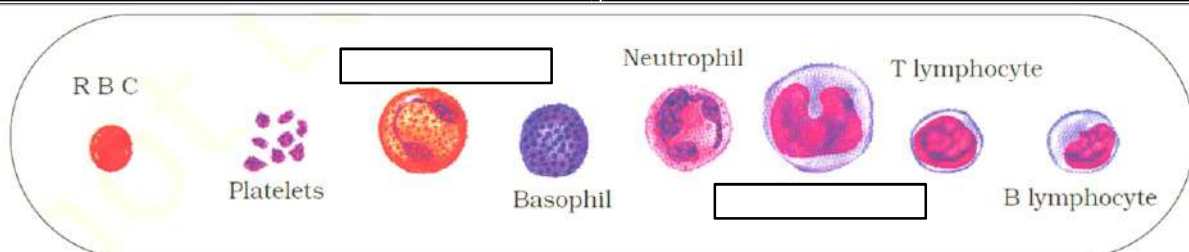


Figure 15.1 Diagrammatic representation of formed elements in blood

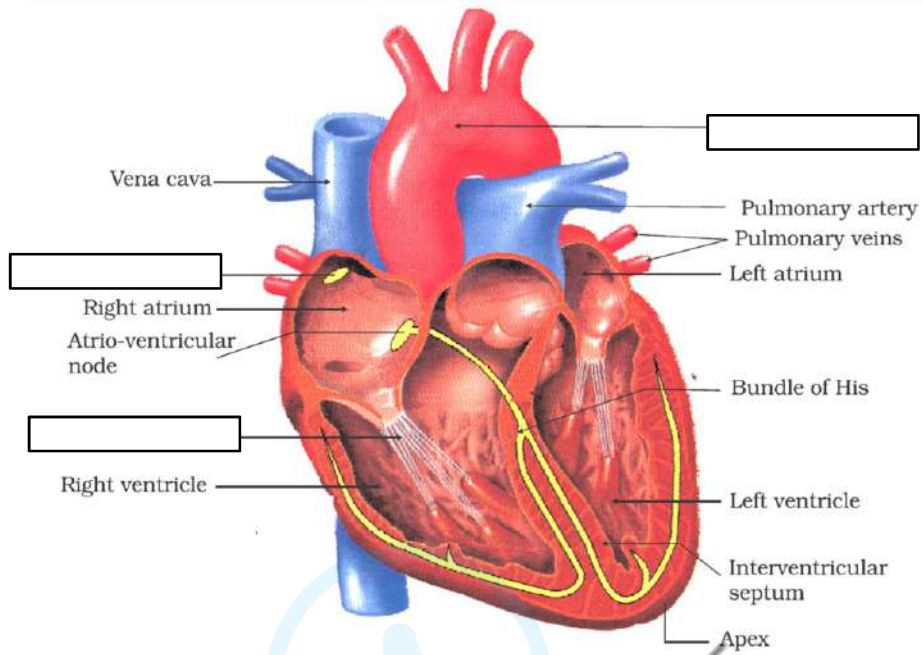


Figure 15.2 Section of a human heart

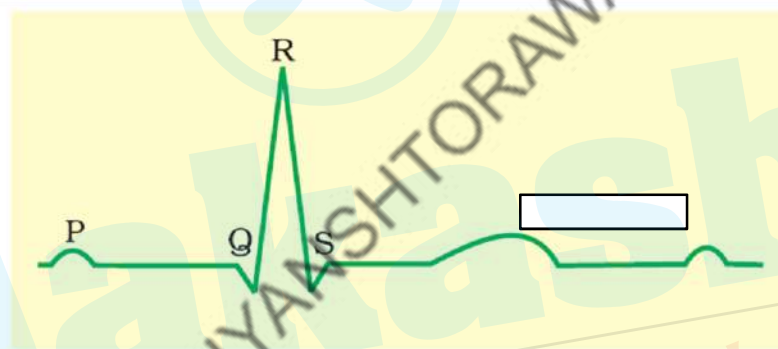


Figure 15.3 Diagrammatic presentation of a standard

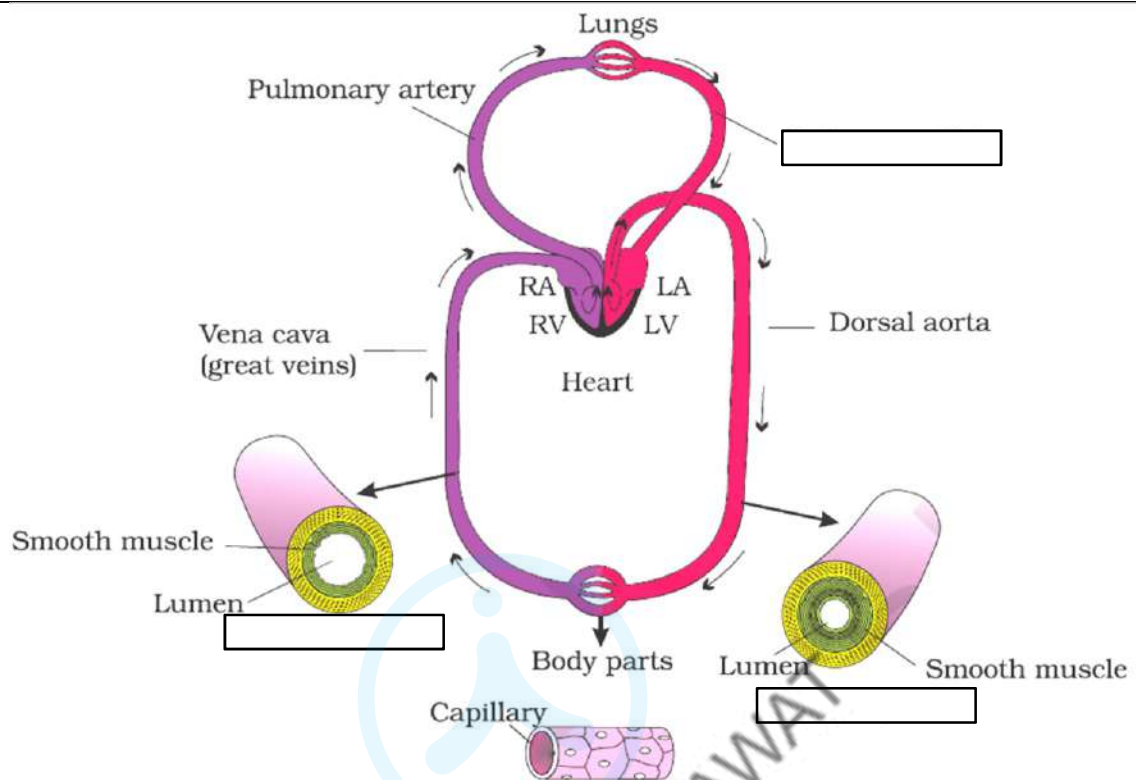


Figure 15.4 Schematic plan of blood circulation in human

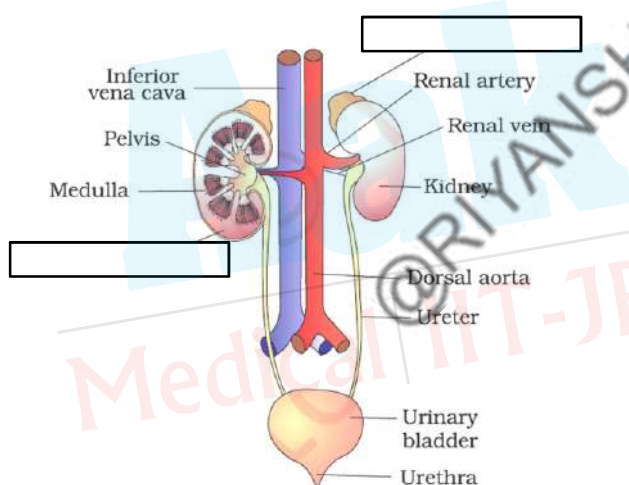


Figure 16.1 Human Urinary system

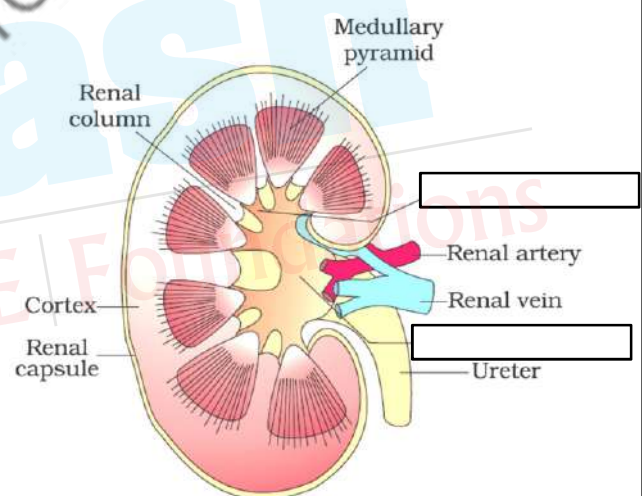


Figure 16.2 Longitudinal section (Diagrammatic) of Kidney

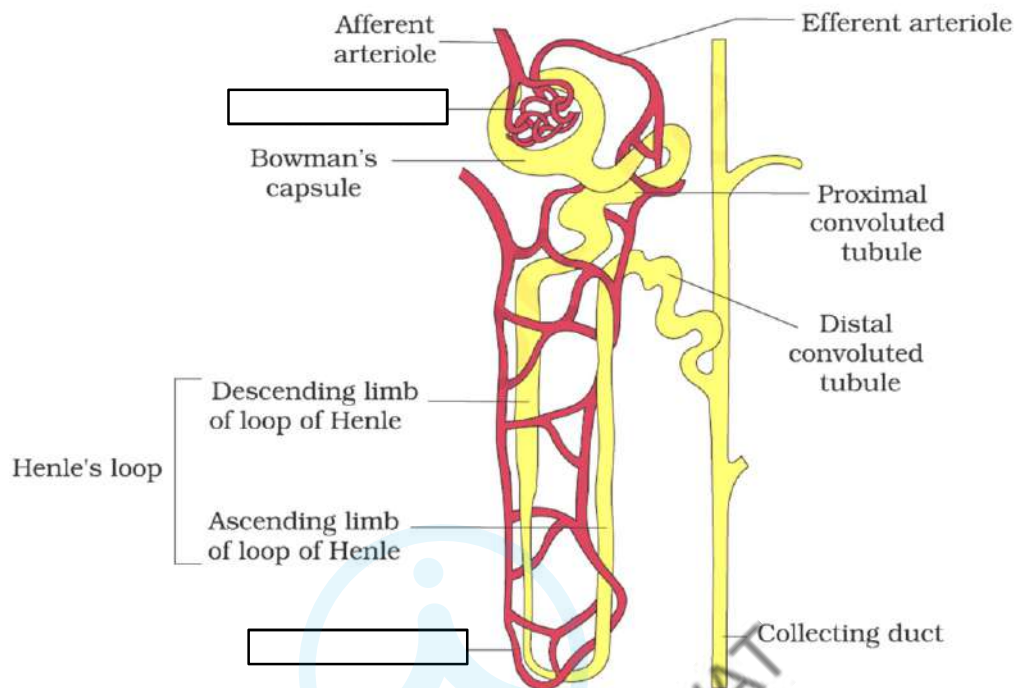


Figure 16.3 A diagrammatic representation of a nephron showing blood vessels, duct and tubule

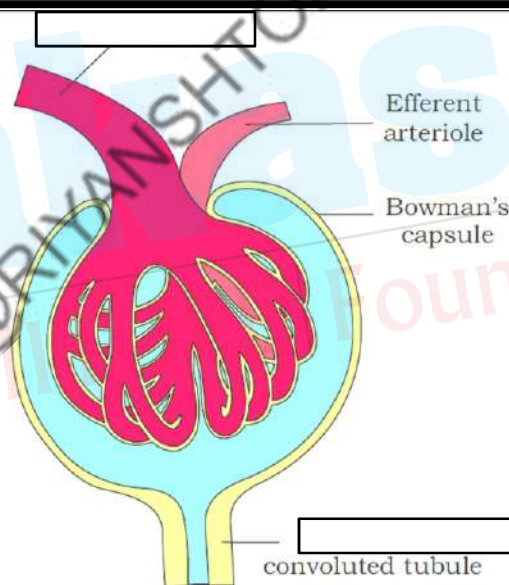


Figure 16.4 Malpighian body (renal corpuscle)

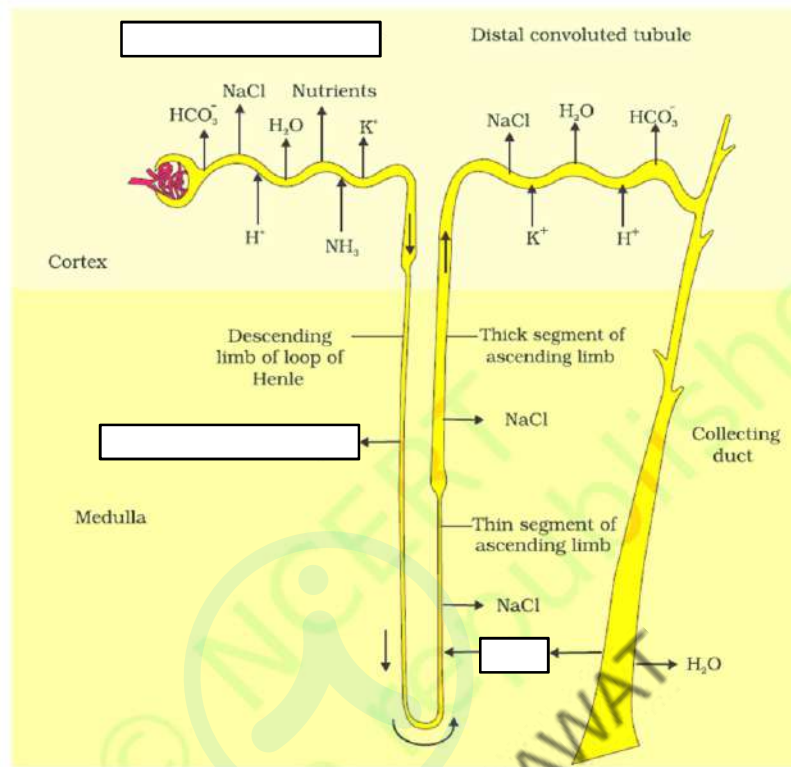


Figure 16.5 Reabsorption and secretion of major substances at different parts of the nephron (Arrows indicate direction of movement of materials.)

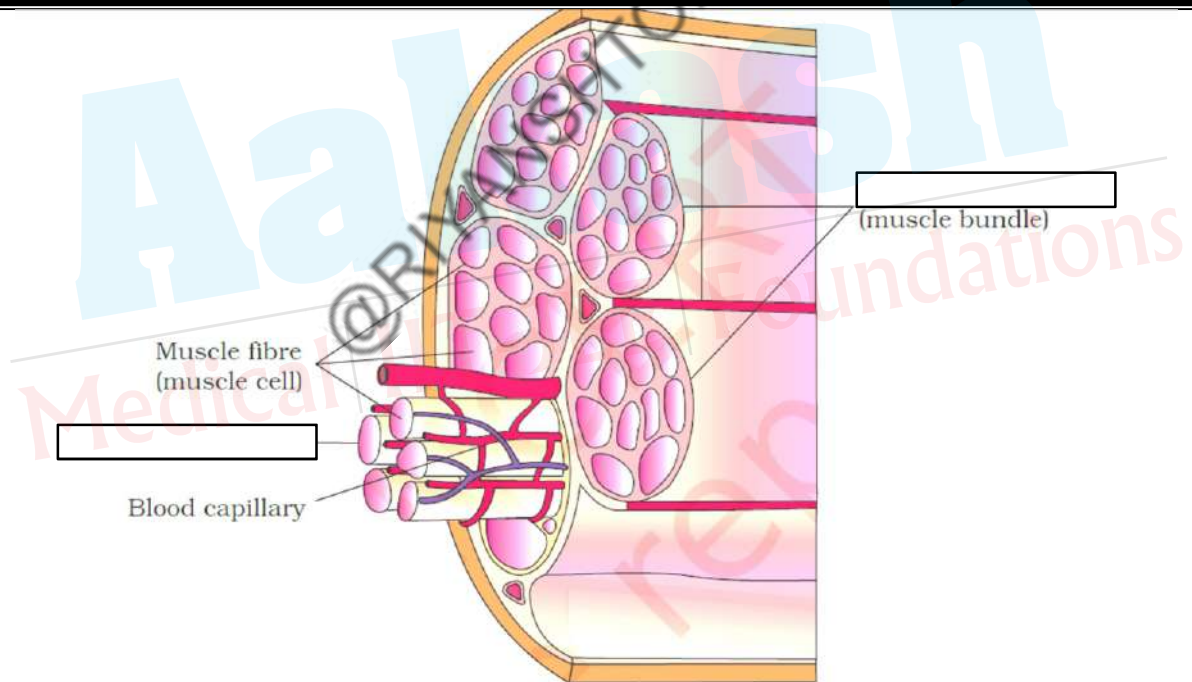


Figure 17.1 Diagrammatic cross sectional view of a muscle showing muscle bundles

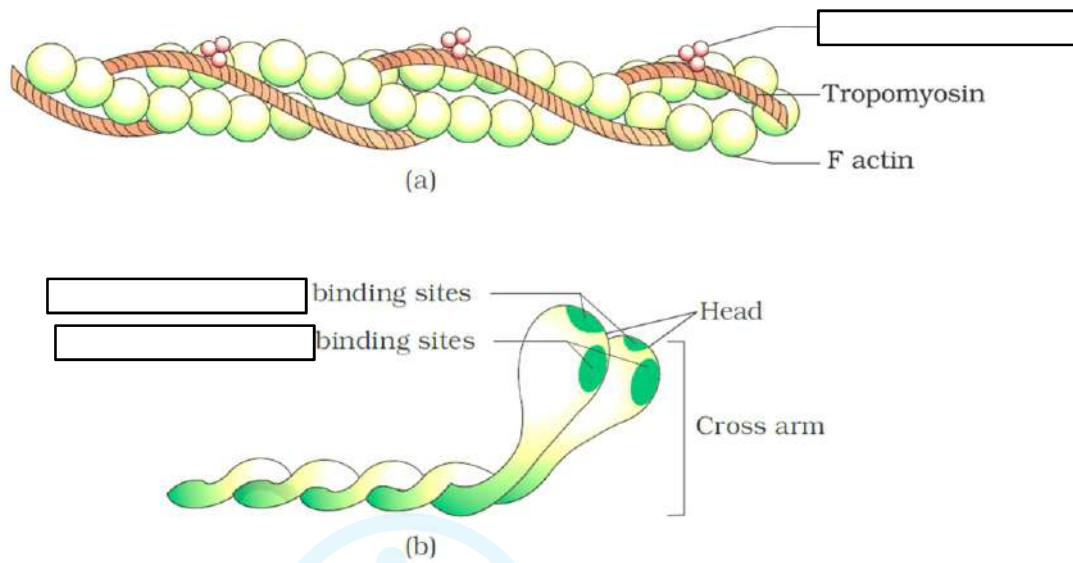


Figure 17.3 (a) An actin (thin) filament (b) Myosin monomer (Meromyosin)

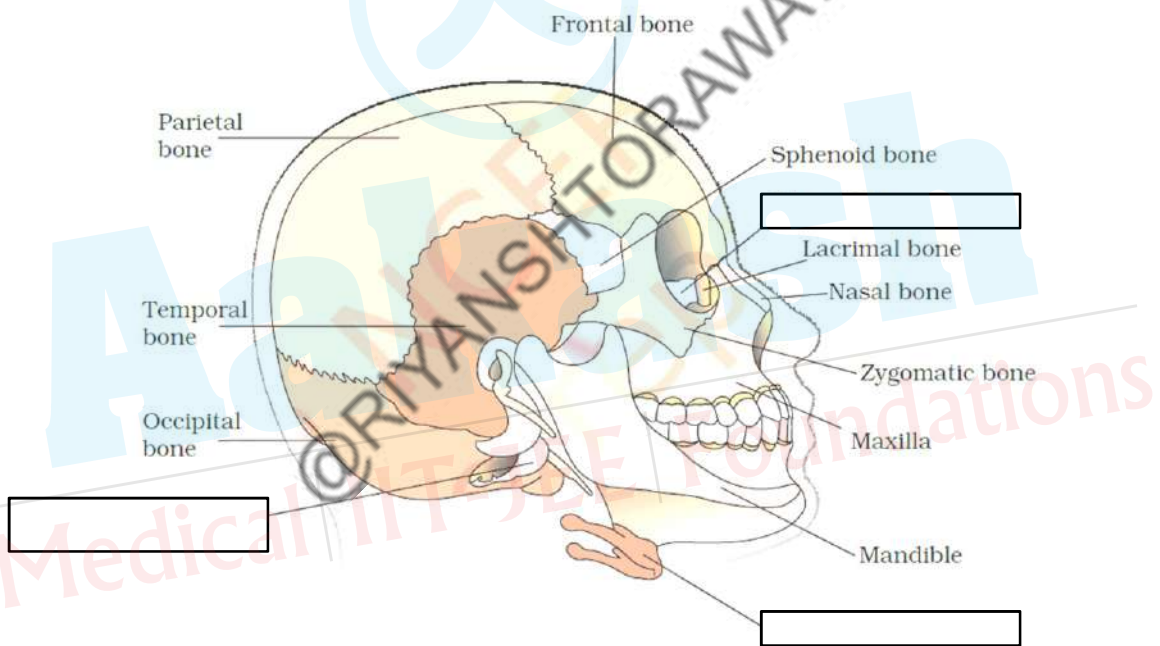


Figure 17.6 Diagrammatic view of human skull

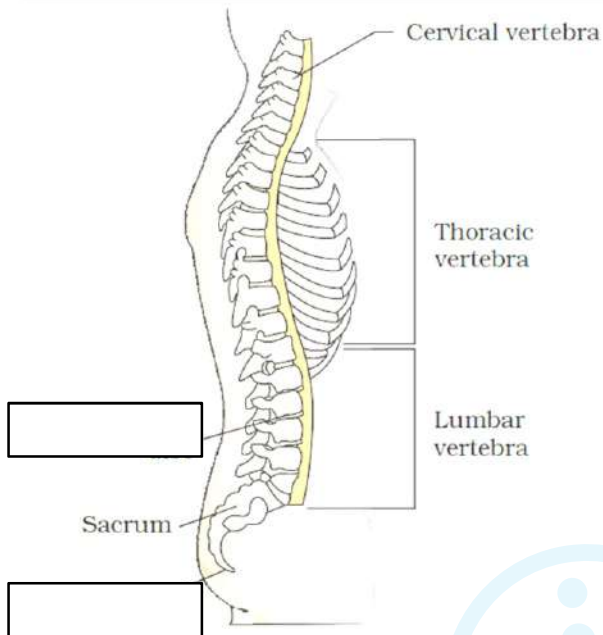


Figure 17.7 Vertebral column (right lateral view)

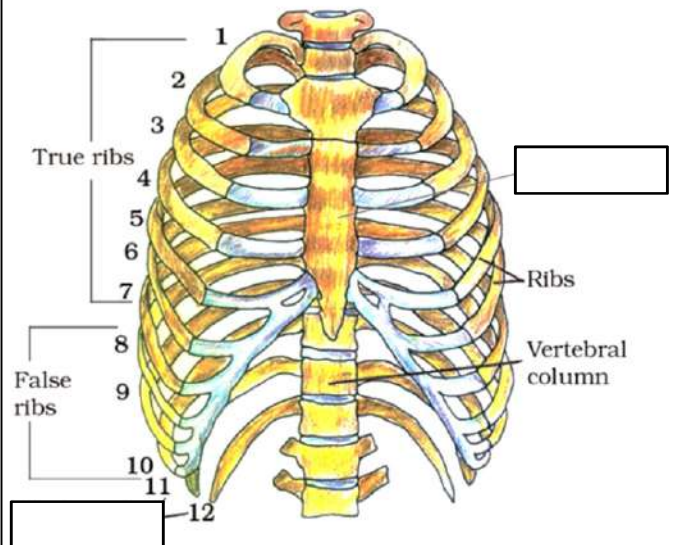


Figure 17.8 Ribs and rib cage

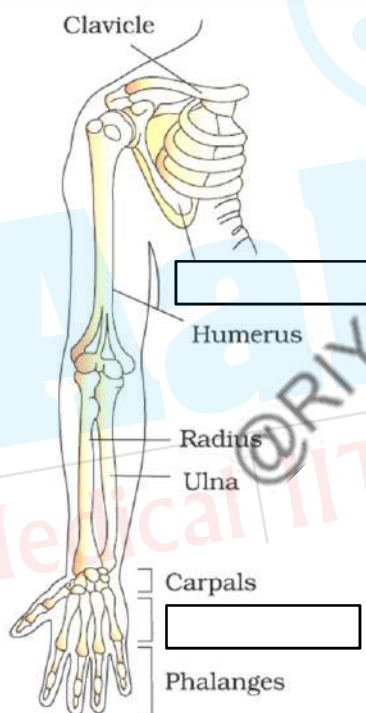


Figure 17.9 Right pectoral girdle and upper arm. (frontal view)

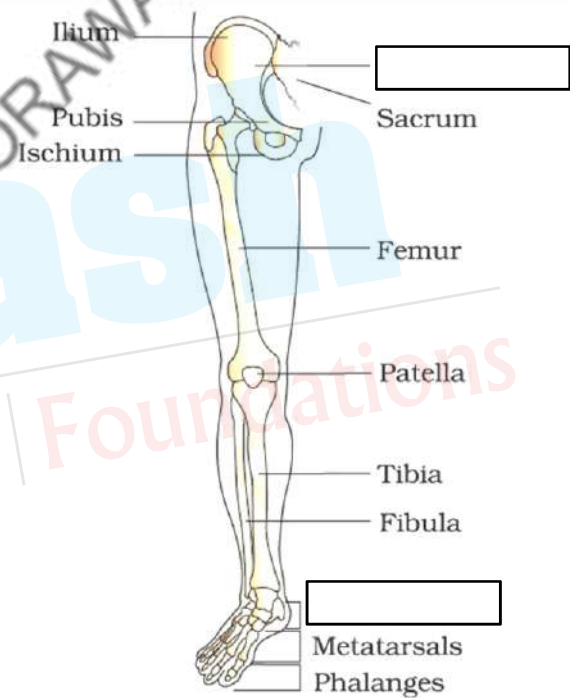


Figure 17.10 Right pelvic girdle and lower limb bones (frontal view)

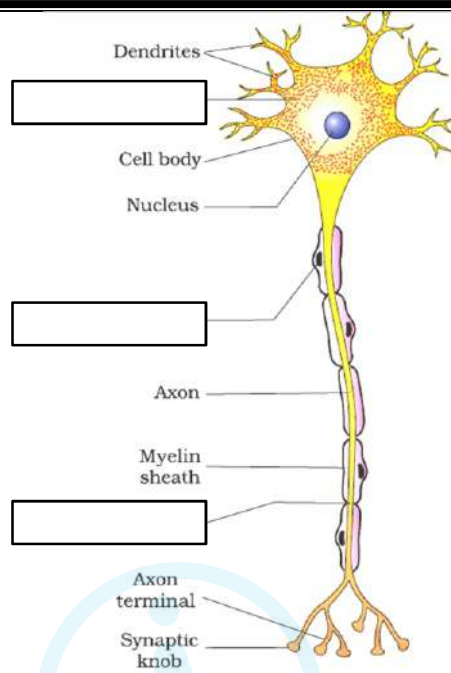


Figure 18.1 Structure of a neuron

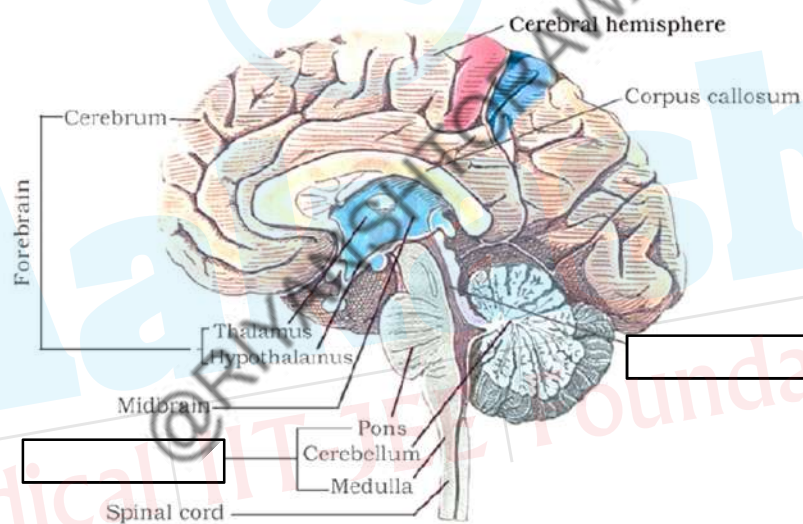


Figure 18.4 Diagram showing sagittal section of the human brain

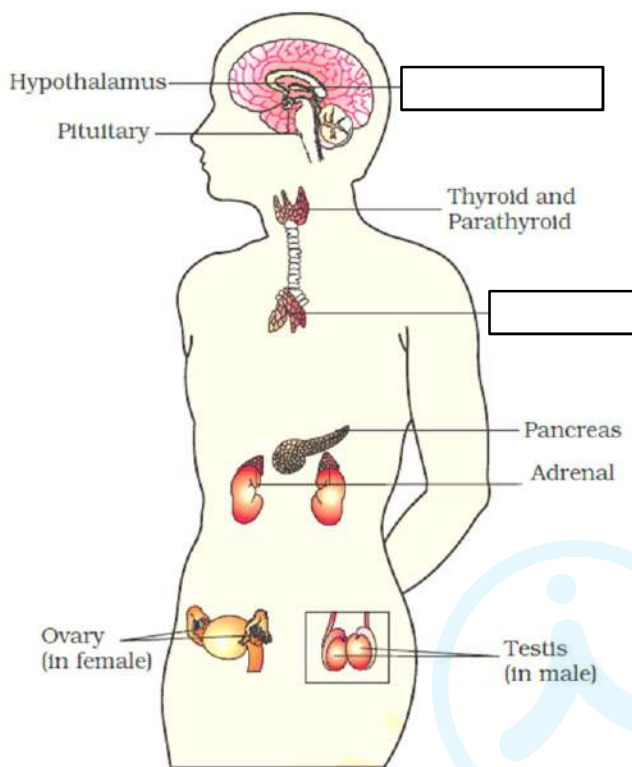


Figure 19.1 Location of endocrine glands

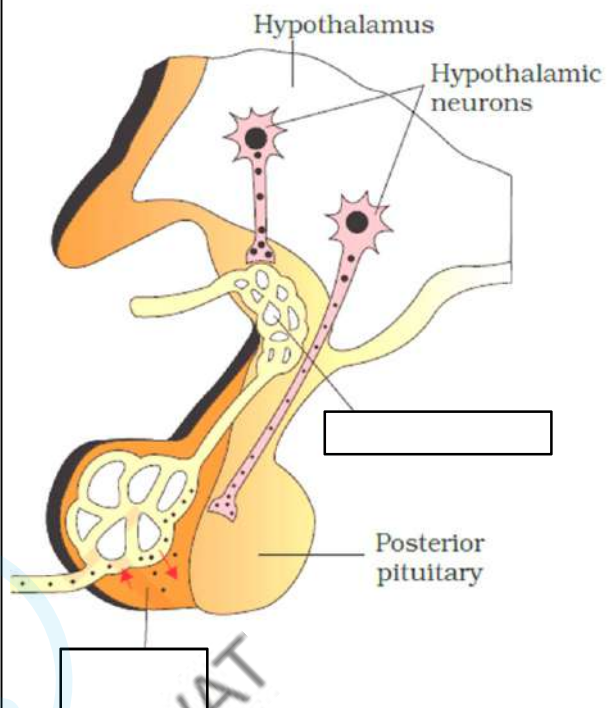
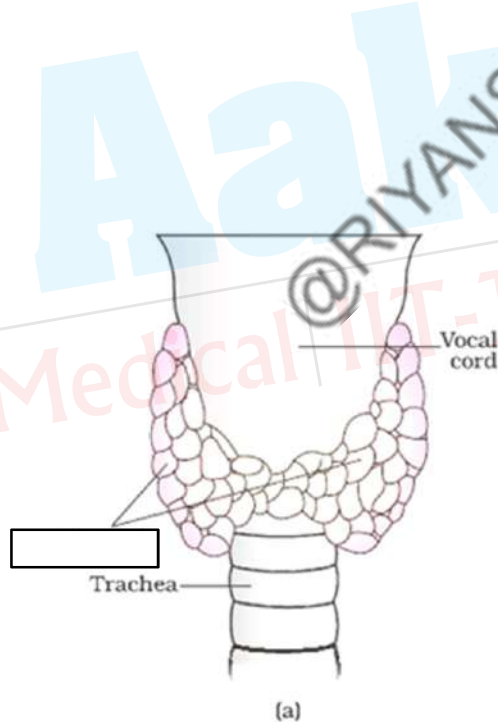


Figure 19.2 Diagrammatic representation of pituitary and its relationship with hypothalamus



(a)

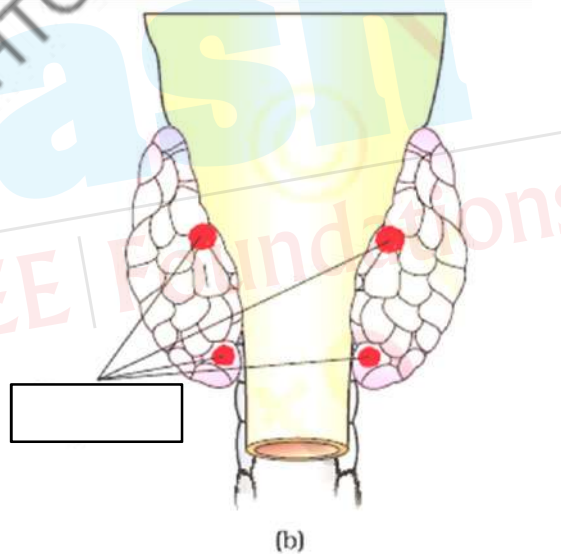


Figure 19.3 Diagrammatic view of the position of Thyroid and Parathyroid
(a) Ventral side
(b) Dorsal side

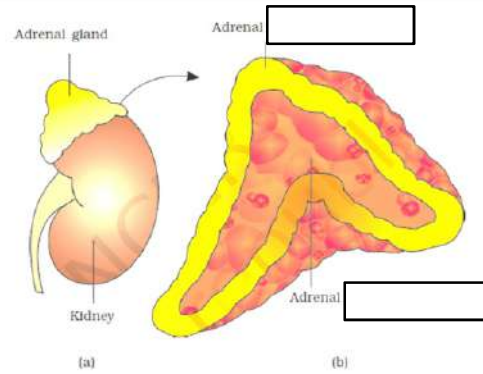


Figure 19.4 Diagrammatic representation of : (a) Adrenal gland above kidney (b) Section showing two parts of adrenal gland

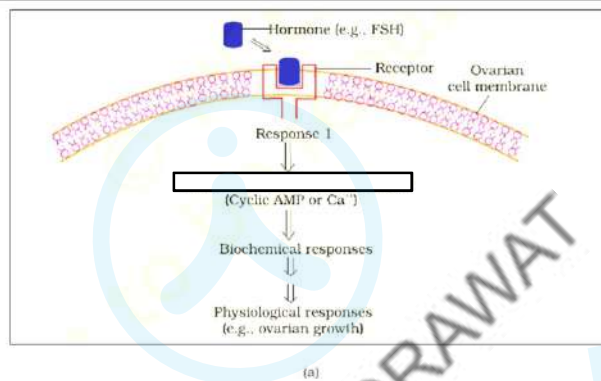


Figure 19.5 Diagrammatic representation of the mechanism of hormone action :
(a) Protein hormone

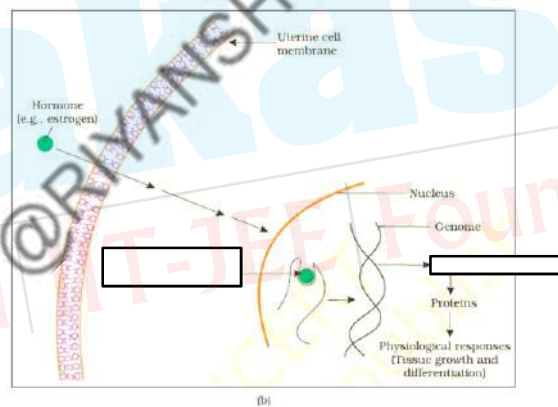


Figure 19.5 Diagrammatic representation of the mechanism of hormone action :
(a) Protein hormone (b) Steroid hormone

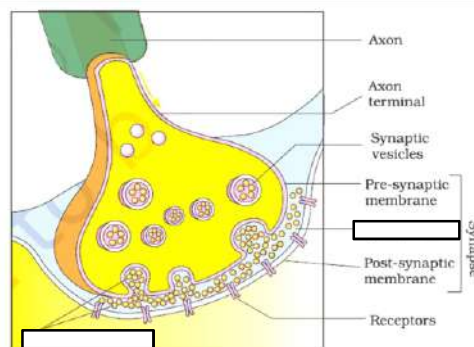


Figure 18.3 Diagram showing axon terminal and synapse



Aakash

Medical | IIT-JEE | Foundations

Regd. Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

Diwali Home Work

Class –XII_ Zoology

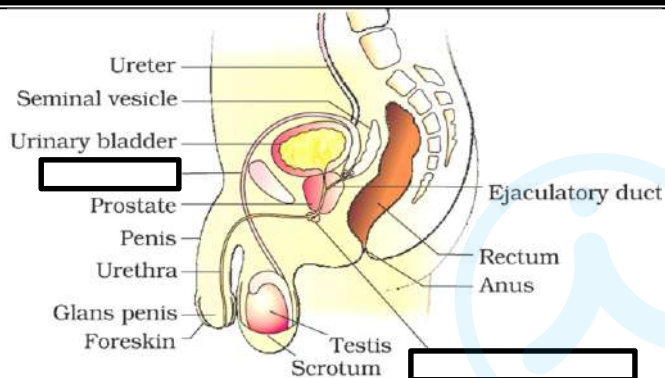


Figure 2.1(a) Diagrammatic sectional view of male pelvis showing reproductive system

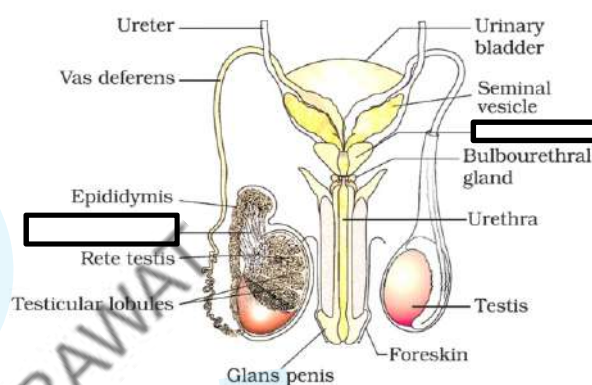


Figure 2.1(b) Diagrammatic view of male reproductive system (part of testis is open to show inner details)

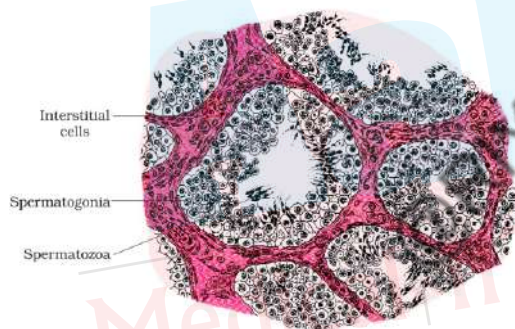


Figure 2.2 Diagrammatic sectional view of seminiferous tubule

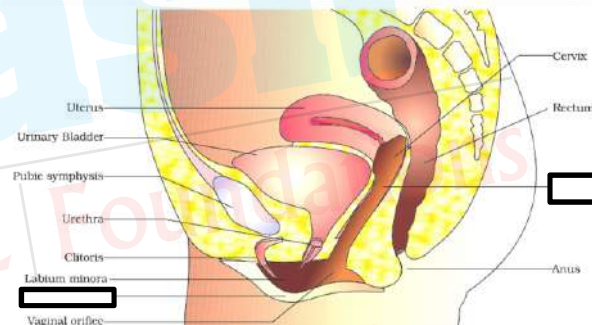


Figure 2.3 (a) Diagrammatic sectional view of female pelvis showing reproductive system

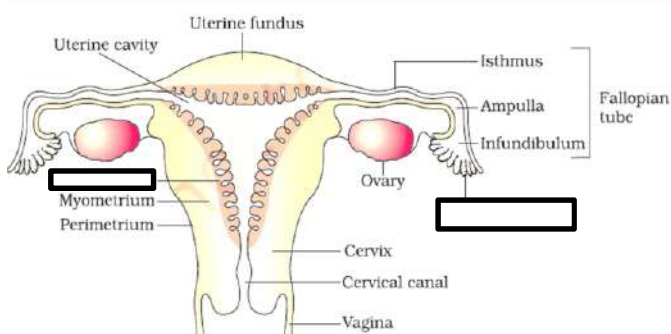


Figure 2.3 (b) Diagrammatic sectional view of the female reproductive system

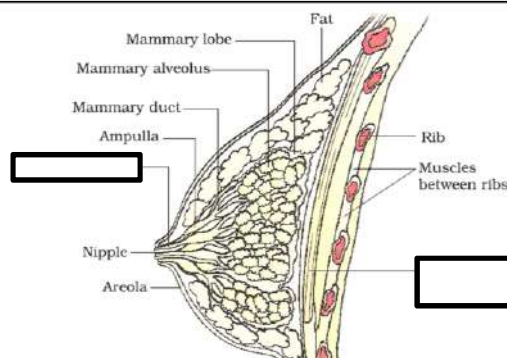


Figure 2.4 A diagrammatic sectional view of Mammary gland

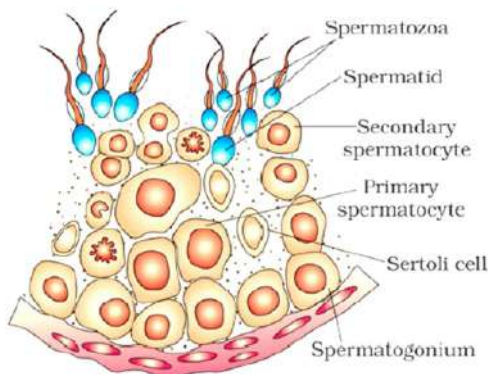


Figure 2.5 Diagrammatic sectional view of a seminiferous tubule (enlarged)

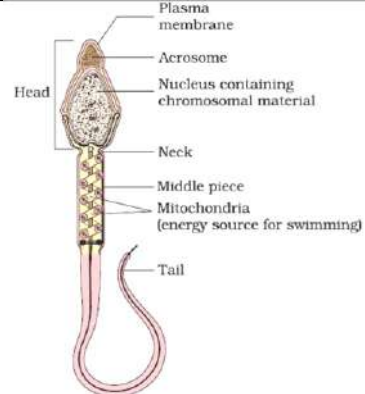


Figure 2.6 Structure of a sperm

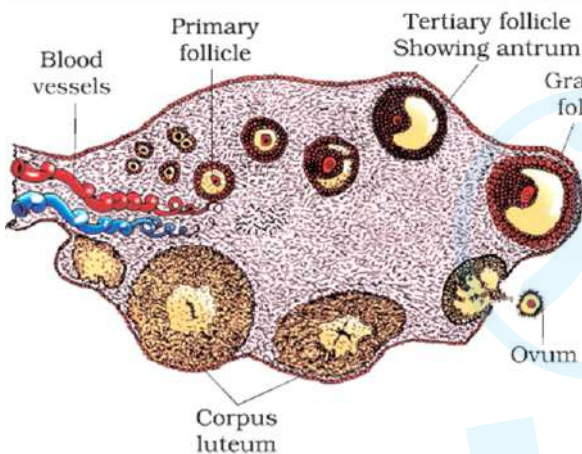


Figure 2.7 Diagrammatic Section view of ovary

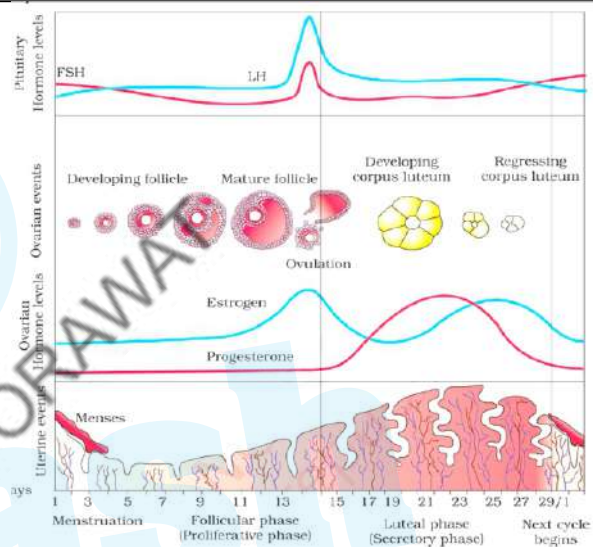


Figure 2.9 Diagrammatic presentation of various events during a menstrual cycle

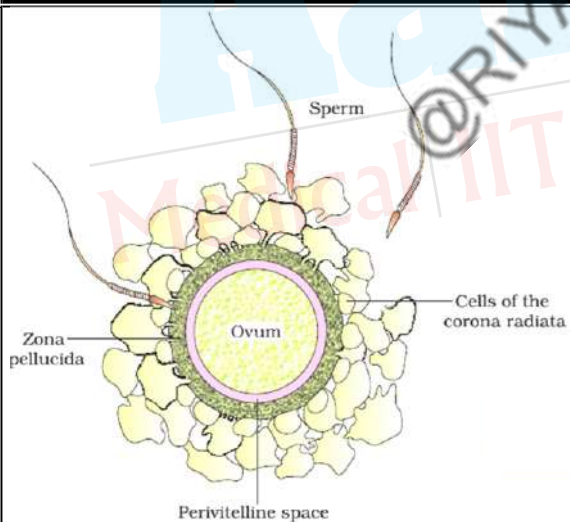


Figure 2.10 Ovum surrounded by few sperms

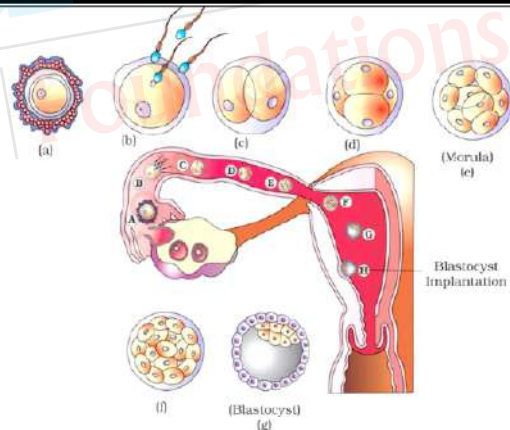


Figure 2.11 Transport of ovum, fertilisation and passage of growing embryo through fallopian tube

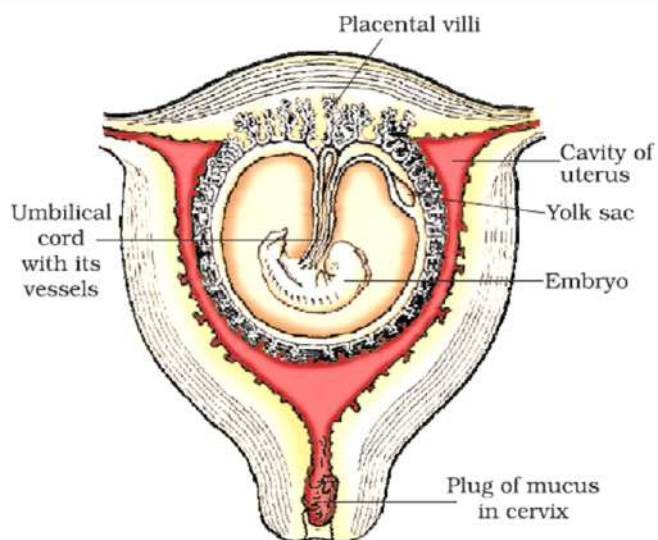


Figure 2.12 The human foetus within the uterus

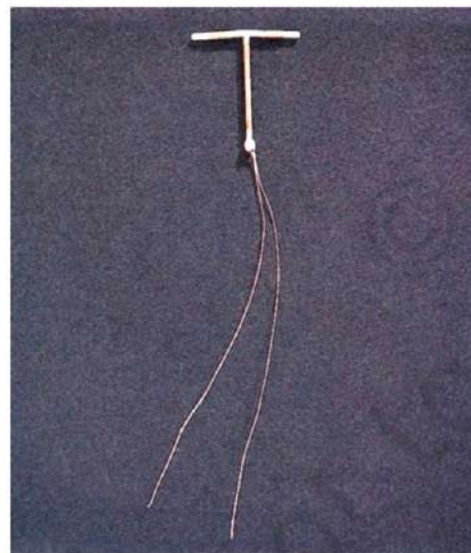


Figure 3.2. Copper T (CuT)



Figure 3.3 Implants

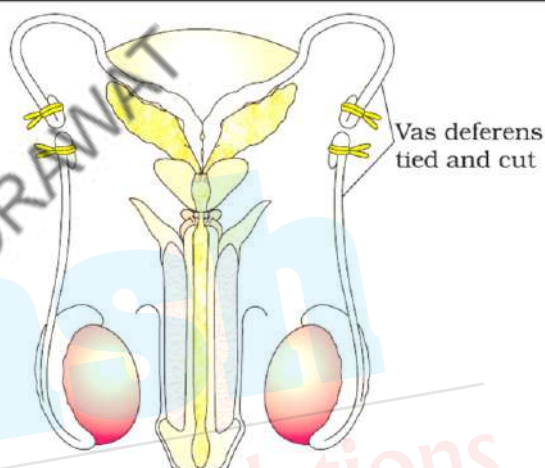


Figure 3.4 (a) Vasectomy

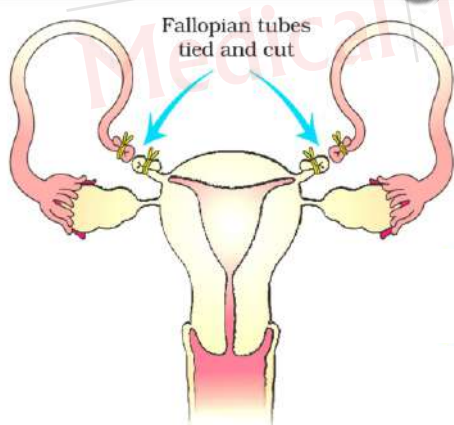


Figure 3.4 (b) Tubectomy

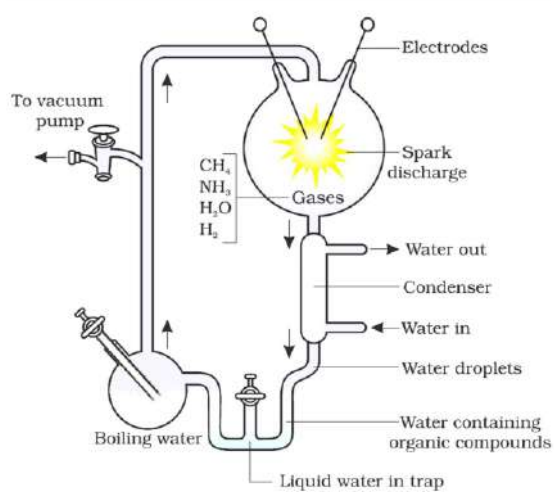


Figure 6.1 Diagrammatic representation of Miller's experiment

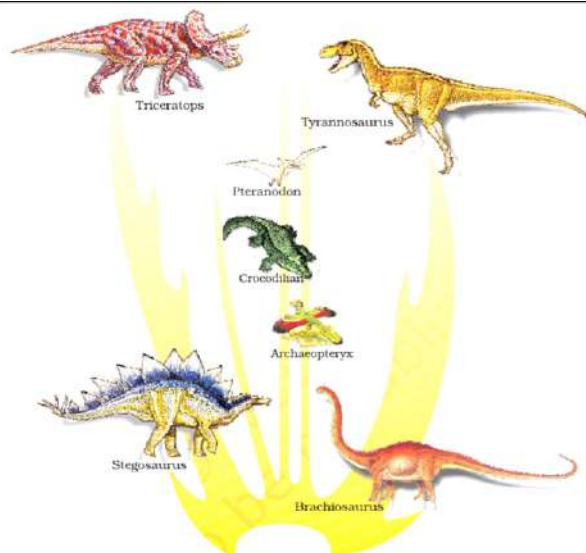


Figure 6.2 A family tree of dinosaurs and their living modern day counterpart organisms like crocodiles and birds

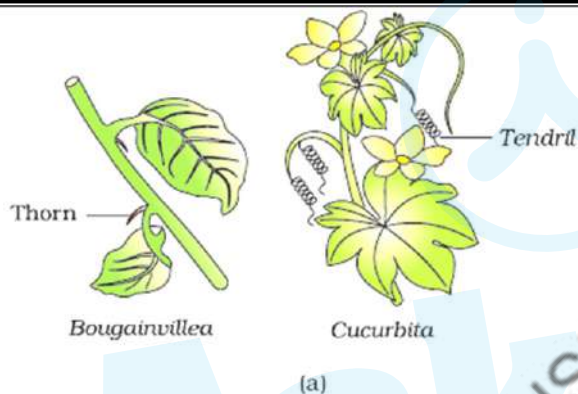


Figure 6.3 Example of homologous organs in (a) Plants and (b) Animals

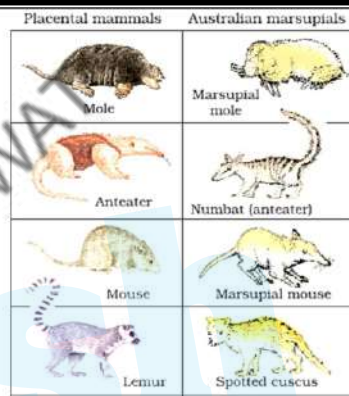


Figure 6.7 Picture showing convergent evolution of Australian Marsupials and placental mammals

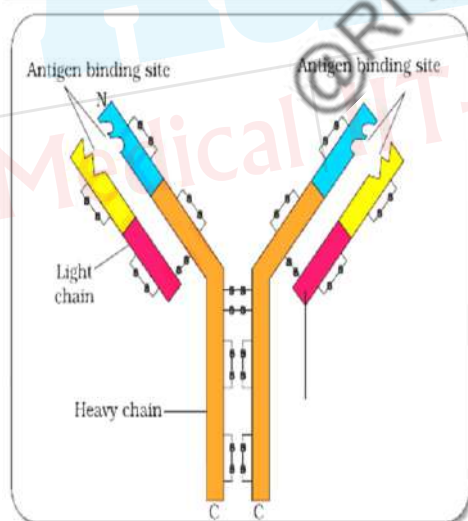


Figure 7.4 Structure of an antibody molecule

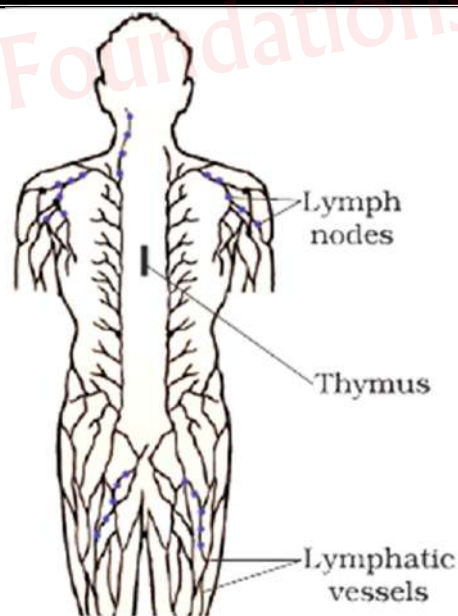


Figure 7.5 Diagrammatic representation of Lymph nodes

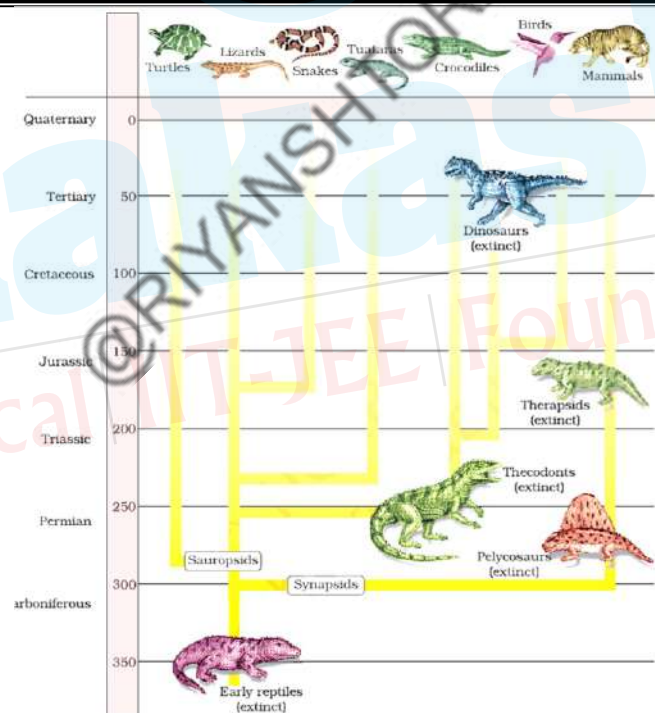
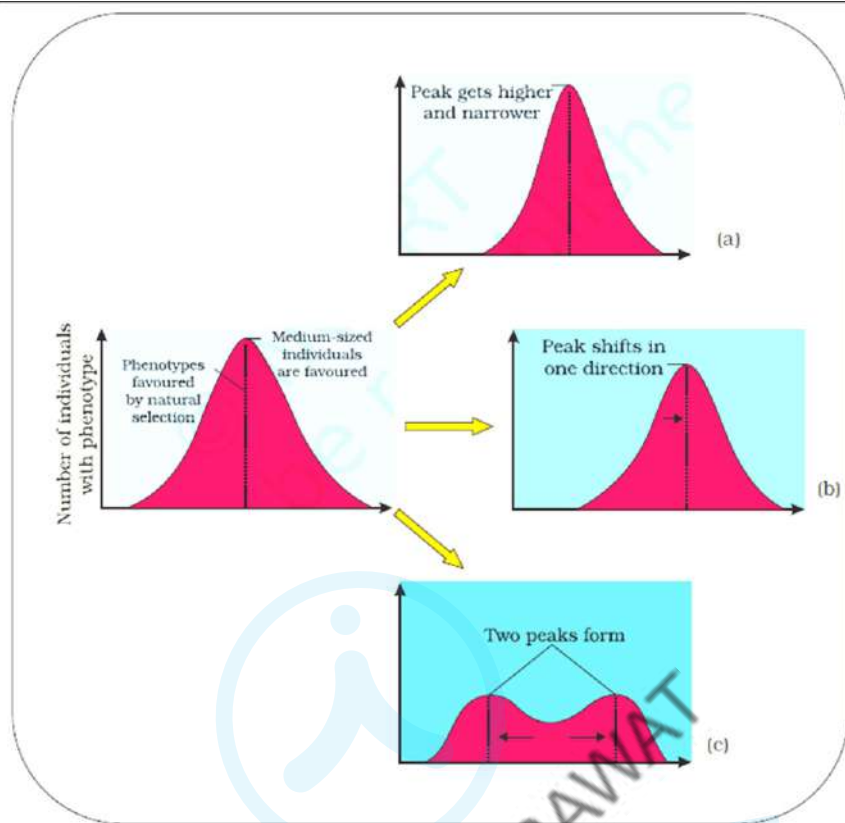


Figure 6.10 Representative evolutionary history of vertebrates through geological periods

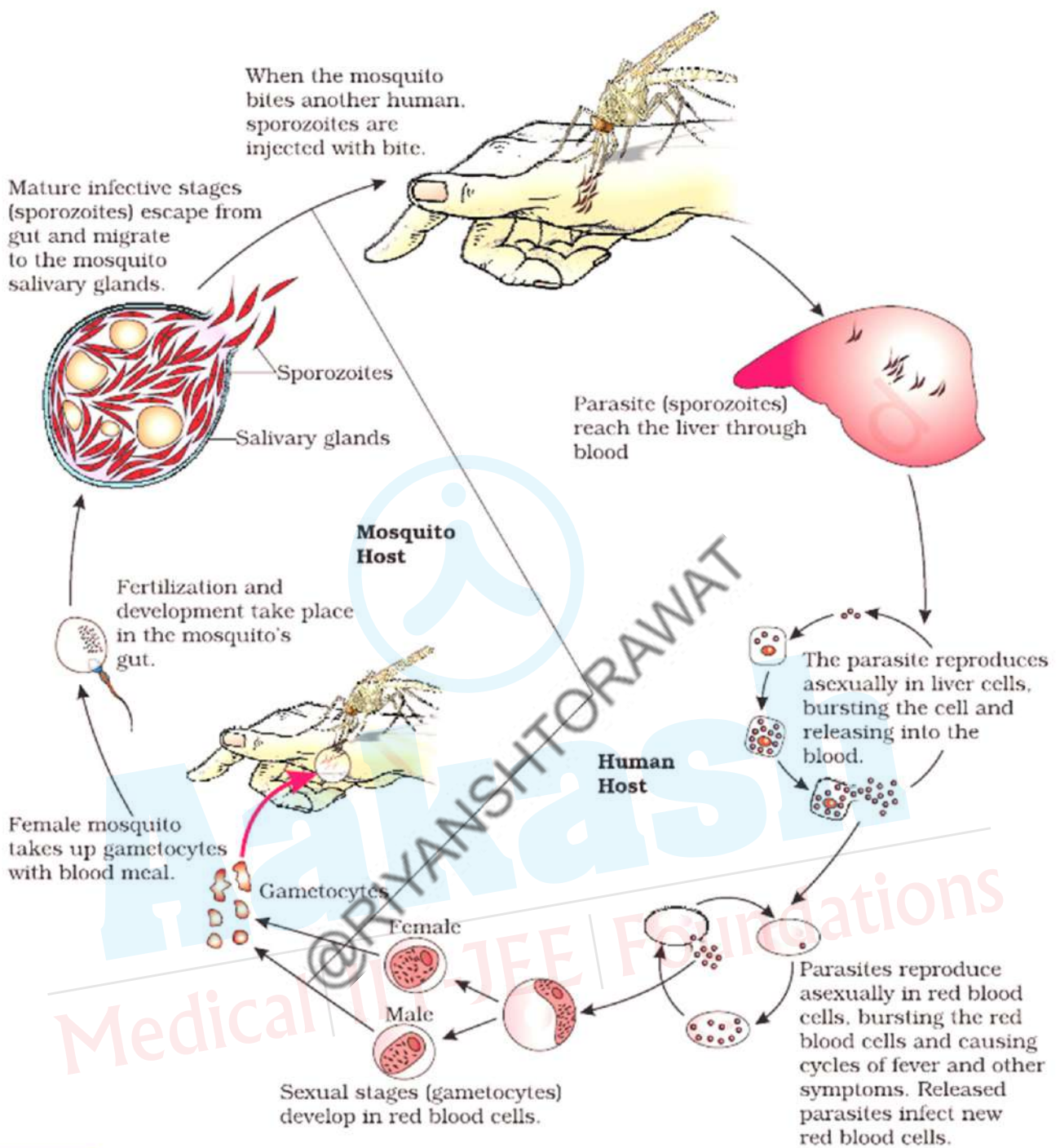


Figure 7.1 Stages in the life cycle of *Plasmodium*

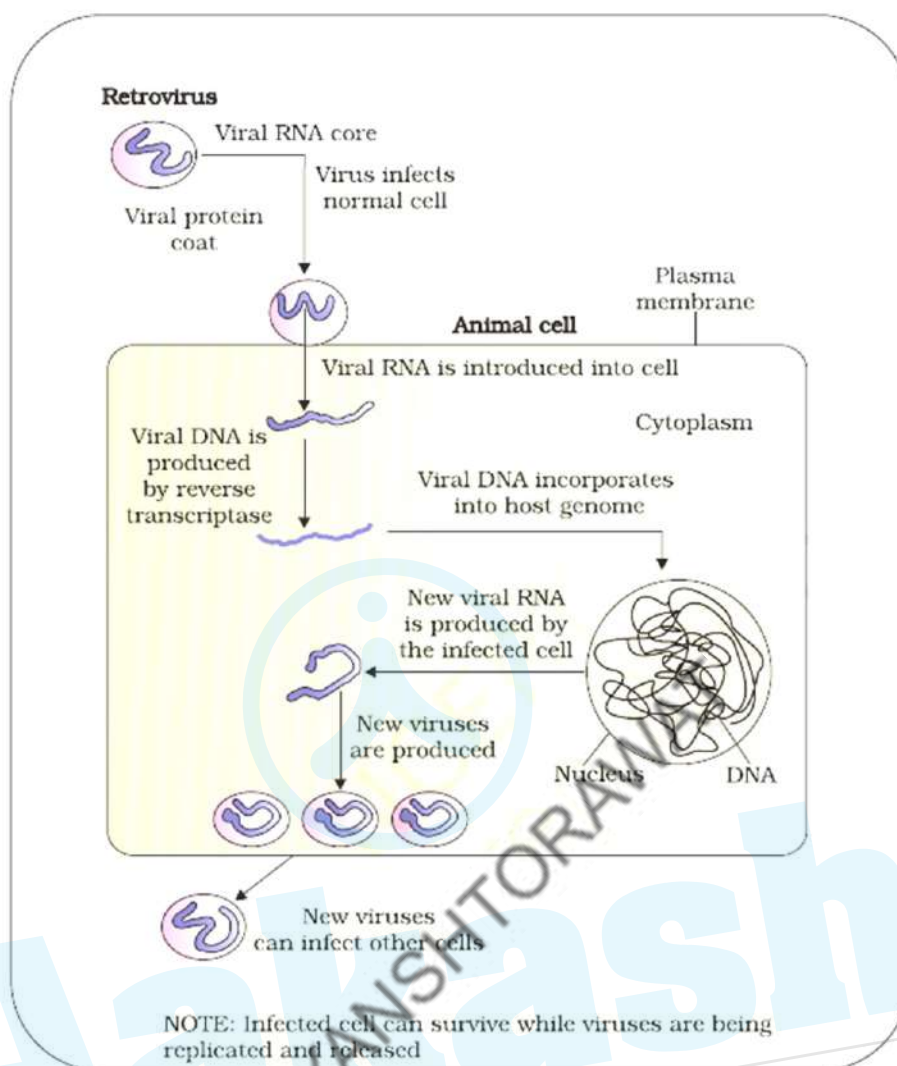


Figure 7.6 Replication of retrovirus

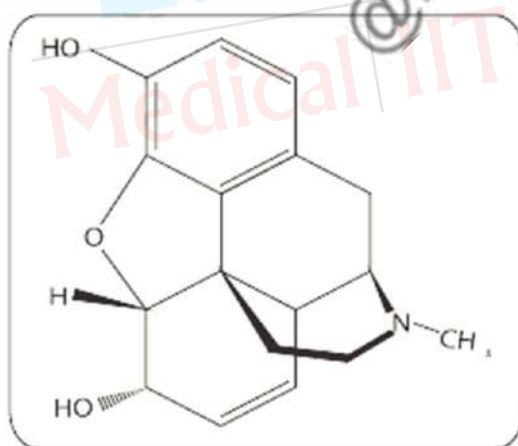


Figure 7.7 Chemical structure of Morphine



Figure 7.8 Optimum poppy

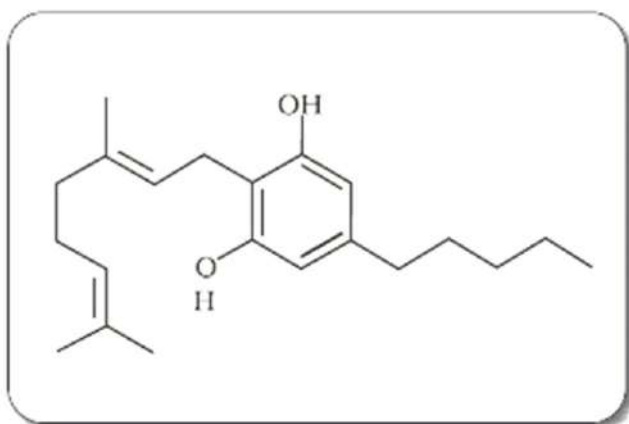


Figure 7.9 Skeletal structure of cannabinoid molecule



Figure 7.10 Leaves of *Cannabis sativa*



Figure 7.11 Flowering branch of *Datura*

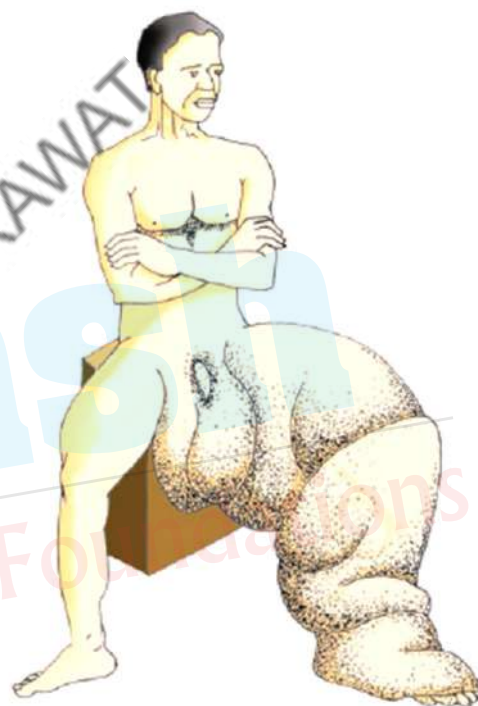


Figure 7.2 Diagram showing inflammation in one of the lower limbs due to elephantiasis